KING MONGKUT'S UNIVERSITY OF TECHNOLOGY THONBURI

Faculty of Engineering

Department of Computer Engineering

FINAL PROJECT REPORT

An Exploration of Artificial Intelligence for Detection of Inhibition Zones in Antibiotic Susceptibility Testing

Submitted by

Peeraphat Siripathanakun Student ID: 65070505214

Phattaradon Thanaprasitpattana Student ID: 65070505223

Advisors

Assoc. Prof. Dr. Nontapon Leeraj

Assoc. Prof. Dr. Panit Ajchararit

Dr. Kritsana Jaengsamsi

Academic Year 2025

ABSTRACT

Antibiotic resistance is a critical global health challenge that demands accurate and rapid diagnostic methods. The Kirby-Bauer disk diffusion method is the internationally accepted standard for Antimicrobial Susceptibility Testing (AST), yet manual measurement of zone of inhibition (ZOI) diameters remains time-consuming and prone to human error. This project develops an artificial intelligence (AI)-based web application to automate the detection and measurement of inhibition zones from photographs of antibiotic susceptibility testing plates.

The system integrates deep learning object detection models—Faster R-CNN with a ResNet-50 backbone and YOLOv11n—together with classical image processing via Hough Circle Transform, to localize and measure ZOI diameters. An Optical Character Recognition (OCR) pipeline using EasyOCR with preprocessing (global thresholding, adaptive binarization, and image rotation) identifies antibiotic disk labels on the plate. Detected zone diameters are interpreted against CLSI 2025 and EUCAST 2023 breakpoints to classify bacteria as Susceptible (S), Intermediate (I), or Resistant (R).

A dataset of 287 real laboratory images was augmented to 543 training images and supplemented with 2,000 synthetic images to improve model generalization. Experimental results demonstrate that both deep learning models outperform Hough Transform in ZOI detection accuracy. YOLOv11n achieved superior accuracy with a mean absolute error (MAE) of approximately **2.5–2.7 mm** and a mAP@0.5 of 95.91%, significantly outperforming the Faster R-CNN baseline (MAE 4.65 mm). The complete system, including the React frontend, FastAPI backend, and SQLite database, is fully integrated and has been **successfully deployed for production use at the Chulabhorn Royal Academy (CRA)** hospital server. User evaluation by five domain experts yielded an overall usefulness score of 3.60 out of 5.00, confirming the system's readiness for real-world microbiological laboratory workflows.

Keywords: Antimicrobial Susceptibility Testing, Zone of Inhibition, Deep Learning, Object Detection, Faster R-CNN, YOLOv11, Hough Transform, EasyOCR, Image Processing, CLSI, EUCAST, Web Application

CONTENTS

Abstract **i**

1 Introduction **1**

1.1 Background and Problem Statement 1

1.2 Motivation 4

1.3 Objectives 5

1.4 Project Scope 5

1.5 Expected Outcomes 6

1.6 Report Structure 7

2 Related Work and Background Theory **8**

2.1 Antibiotic Susceptibility Testing 8

2.1.1 Clinical Standards: CLSI and EUCAST 9

2.1.2 Clinical Significance of Measurement Precision 10

2.2 Deep Learning for Object Detection 10

2.2.1 Faster R-CNN 11

2.2.2 YOLOv11 12

2.2.3 Hough Circle Transform 14

2.3 Optical Character Recognition for Antibiotic Disk Labeling 15

2.3.1 EasyOCR and the CRAFT Architecture 15

2.4 Image Preprocessing Techniques 16

2.5 Related Work in Automated AST Analysis 17

2.5.1 Deep Learning Approaches for AST Image Analysis 17

2.5.2 Evaluation Standards for Automated AST Methods 18

3 Methodology **20**

3.1 System Architecture Overview 20

3.2 Dataset Collection and Preparation 22

3.2.1 Real Image Collection 22

3.2.2 Data Augmentation 22
3.2.3 Synthetic Data Generation 23
3.2.4 Final Dataset Split 24
3.3 Annotation 24
3.4 Model Training 25
3.4.1 Faster R-CNN 25
3.4.2 YOLOv11n 26
3.4.3 Hough Circle Transform 26
3.5 ZOI Diameter Measurement and Calibration 27
3.6 OCR Pipeline for Antibiotic Label Recognition 28
3.6.1 Disk Crop and Preprocessing Strategy 28
3.7 System Deployment and Infrastructure 29
3.7.1 Server Environment and Backend Services 29
3.7.2 Reverse Proxy and Security 30
3.7.3 Frontend Serving 30
3.7.4 Post-Processing and Antibiotic Name Matching 30
3.8 Susceptibility Classification 31

4 Results and Discussion 33
4.1 Object Detection Performance 33
4.1.1 Per-Antibiotic Error Analysis 34
4.1.2 YOLOv11n vs. Faster R-CNN: Architectural Explanation . . . 36
4.1.3 Hough Transform Baseline 37
4.1.4 Detection Accuracy (mAP) 38
4.2 Effect of Dataset Size on Training 38
4.2.1 Underfitting at Very Small Dataset Sizes (54 and 274 Images) 39
4.2.2 Overfitting Onset at Intermediate Size (702 Images) 39
4.2.3 Optimal Generalization at Full Dataset (1,884 Images) . . . 39
4.3 OCR Performance 40
4.3.1 Preprocessing Pipeline 41
4.3.2 Comparison Results 42
4.4 Calibration and Pixel-to-mm Conversion 43

4.4.1 The Calibration Problem 43
4.4.2 Calibration Formula 43
4.4.3 Initial vs. Refined Calibration 44
4.5 Processing Time 44
4.5.1 End-to-End Pipeline Timing 44
4.5.2 Clinical Workflow Context 45
4.6 User Evaluation 45
4.6.1 Participant Profile 45
4.6.2 Scenario-Based Test Results 46
4.6.3 Usability Ratings (Likert Scale 1–5) 47
4.6.4 Discussion of Accuracy Trust Score (3.20/5.00) 47
4.6.5 Overall Evaluation Summary 48
4.7 Web Application 48

5 Conclusion and Future Work 50
5.1 Conclusion 50
5.1.1 Summary of the Project 50
5.1.2 Quantitative Summary 51
5.1.3 Project Completion Status 51
5.2 Limitations 52
5.2.1 Residual Calibration Bias 52
5.2.2 Dataset Diversity 52
5.3 Future Work 52
5.3.1 Large-Scale Clinical Validation Study 52
5.3.2 Instance Segmentation for Ellipse Fitting 53
5.3.3 Mobile and On-Device Deployment 53
5.3.4 Clinical Validation Study 53
5.3.5 Epidemiology and Resistance Trend Module 53

A แบบสอบถามการประเมินระบบ 60

LIST OF FIGURES

1.1 Representative Kirby-Bauer disk diffusion plate illustrating zones of inhibition (ZOI) around antibiotic-impregnated disks. The diameter of each clearing zone is measured to determine antibiotic susceptibility. 2

2.1 Architecture of Faster R-CNN, showing the shared convolutional backbone, the Region Proposal Network (RPN) producing candidate object regions, and the detection head that classifies and refines each proposal. 11

2.2 Schematic architecture of YOLOv11, illustrating the backbone feature extraction network, the feature pyramid neck for multi-scale detection, and the decoupled detection heads for bounding box regression and class prediction. 13

2.3 Illustration of the Hough Circle Transform. Each edge point votes for a cone of candidate circles in the (x_c, y_c, r) parameter space; genuine circles produce prominent peaks in the accumulator. 14

3.1 End-to-end image analysis pipeline of the automated AST system, from image input through detection, OCR, calibration, and breakpoint-based susceptibility classification. 21

3.2 Overall system architecture of the automated AST web application, showing the React frontend, FastAPI backend, AI inference pipeline, and SQLite database layer. 21

4.1 Example ZOI detection output from YOLOv11n showing predicted bounding boxes (green) overlaid on an agar plate image. Antibiotic labels identified by OCR are annotated above each bounding box, and the predicted ZOI diameter in mm is shown below. 36

4.2 Mean Absolute Error (MAE) comparison across detection methods (Term 1 results). Faster R-CNN diameter-based MAE: 4.6525 mm; area-based MAE: 4.845 mm. YOLOv11n MAE was not reported in Term 1 and is omitted from this chart. 37

4.3 Training and validation loss curves across dataset sizes (54, 274, 702, 1884 images). Solid lines represent training loss; dashed lines represent validation loss. The convergence gap narrows substantially as dataset size increases. 40

4.4 OCR preprocessing pipeline applied to a Kanamycin (CN) disk crop. (a) Original RGB crop with specular highlight. (b) After CLAHE enhancement — local contrast improved. (c) After Otsu binarization — label characters isolated in black on white background. 42

4.5 Mean user satisfaction scores across usability criteria (n=5). 47

4.6 Web application dashboard showing ZOI measurements and S/I/R classification results. Each row corresponds to one detected antibiotic disk. The “Edit” column allows manual correction of individual measurements before report export. 49

LIST OF TABLES

3.1 Dataset composition and train/validation/test split used for model training and evaluation. 24

4.1 ZOI Measurement Results — Per-Antibiotic Comparison (all values in mm). Expert GT is the single consensus diameter from the clinical expert. "Diam." columns use the bounding-box width directly; "Area" columns derive the effective diameter from the bounding-box area. Error = Model − Expert GT (positive = overestimation). YOLOv11n demonstrates superior accuracy with a mean absolute error (MAE) of approximately 2.5–2.7 mm. 34

4.2 Detection accuracy metrics (mAP@0.5, Precision, Recall) for both models. Results are from the Term 1 evaluation on the held-out test partition. 38

4.3 Validation loss at convergence for each dataset size subset. Lower values indicate better generalization. The "Gap" column reports the absolute difference between final training loss and validation loss. . . 40

4.4 OCR performance comparison: EasyOCR vs. TyphoonOCR under different preprocessing. "Correct Labels" reports the number of correctly identified disk labels out of four total test labels (P, OX, C, CN). 42

4.5 Per-image processing time breakdown for each pipeline stage. Times are reported as mean ± standard deviation across 30 test images. All model inference times use GPU acceleration. 45

4.6 Profiles of user evaluation participants. 46

4.7 Scenario-based test success rates (n=5). A task is counted as "completed" if the participant finished it without assistance within the allotted time. 46

4.8 User satisfaction scores (1 = strongly disagree, 5 = strongly agree, n=5). 47

CHAPTER 1: INTRODUCTION

1.1 Background and Problem Statement

Antimicrobial resistance (AMR) has emerged as one of the most pressing and complex threats to global public health in the twenty-first century. When bacteria, viruses, fungi, or parasites evolve mechanisms to withstand the drugs designed to eliminate them, standard treatments fail and infections persist, spread, and increasingly result in death. The World Health Organization (WHO) has classified AMR as a critical global priority, estimating that drug-resistant infections directly caused approximately 1.27 million deaths in 2019 alone, with indirect contributions to nearly 5 million additional fatalities worldwide [1]. Without coordinated action, it has been projected that AMR could claim up to 10 million lives per year by 2050, surpassing cancer as a leading cause of mortality globally. Beyond the human toll, AMR imposes staggering economic costs through extended hospital stays, more expensive second- and third-line therapies, and reduced workforce productivity.

At the core of combating AMR lies the practice of Antimicrobial Susceptibility Testing (AST). Before initiating antibiotic therapy, clinicians must determine which drugs remain effective against the specific bacterial pathogen infecting a patient. Prescribing an antibiotic to which the organism is already resistant not only fails the patient but also selects for further resistance in the population, perpetuating the crisis. Accurate and timely AST is therefore not merely a diagnostic procedure—it is a public health intervention. Delays in receiving susceptibility results, which can stretch from 24 to 72 hours using culture-based methods, force clinicians to begin empirical broad-spectrum therapy, a practice that itself accelerates resistance development.

The Kirby-Bauer disk diffusion method, first systematically described and validated in the 1960s, remains the most widely adopted technique for AST in clinical and reference microbiology laboratories worldwide [2,3]. Its enduring popularity stems from its simplicity, low cost, and reliability when performed correctly. In this method, a Mueller-Hinton agar plate is inoculated with a standardized suspension of the target bacterium (typically adjusted to a 0.5 McFarland turbidity standard). Commercially manufactured paper disks, each impregnated with a precise mass of a specific antibiotic, are placed

on the inoculated agar surface at defined inter-disk distances to prevent zone overlap. After an overnight incubation period at 35–37°C, the antibiotic diffuses radially outward from each disk, establishing a concentration gradient in the agar. Where the local antibiotic concentration exceeds the minimum inhibitory concentration (MIC) of the bacterium, bacterial growth is suppressed, resulting in a visible circular clearing called the zone of inhibition (ZOI). Where the concentration falls below the MIC, bacterial colonies grow normally.

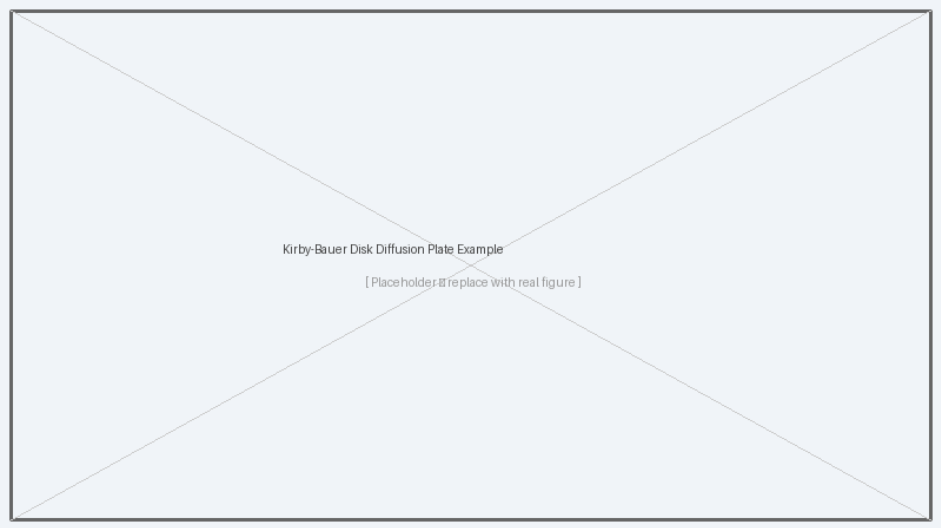

Figure 1.1: Representative Kirby-Bauer disk diffusion plate illustrating zones of inhibition (ZOI) around antibiotic-impregnated disks. The diameter of each clearing zone is measured to determine antibiotic susceptibility.

The diameter of each ZOI is then measured and compared against published interpretive breakpoints to classify the tested organism as Susceptible (S), Intermediate (I), or Resistant (R). These breakpoints are rigorously maintained and updated by two major international bodies: the Clinical and Laboratory Standards Institute (CLSI) in the United States, and the European Committee on Antimicrobial Susceptibility Testing (EUCAST) in Europe [4]. CLSI publishes the M02 standard governing disk diffusion testing procedures, while EUCAST publishes annually revised breakpoint tables derived from pharmacokinetic/pharmacodynamic (PK/PD) modeling, epidemiological cutoff values (ECOFFs), and clinical outcome data [5]. Both organizations provide species- and drug-specific diameter thresholds that must be applied correctly to yield clinically meaningful results.

Despite the apparent simplicity of measuring a circle diameter, manual ZOI measurement is considerably error-prone in practice. Laboratory technicians use rulers, Vernier calipers, or specialized template devices to measure each zone, and inter-observer

variability is a recognized and persistent challenge in the field [6]. Factors such as partially overlapping zones between adjacent disks, heterogeneous or "swarming" growth within the zone boundary, uneven agar surface, irregular plate inoculation, sub-optimal incubation, and inconsistent photographic or visual angle all contribute to measurement variability. Even a single-millimeter discrepancy can be clinically decisive: for example, CLSI 2025 defines a ZOI diameter of 18 mm or above for *Staphylococcus epidermidis* tested against oxacillin (1 μg) as Susceptible, while 17 mm or below is classified as Resistant. With no intermediate category for this drug-organism combination, every millimeter matters. Furthermore, laboratories processing high case volumes require technicians to measure dozens of zones per shift, introducing fatigue as a compounding source of error.

Existing automated AST platforms, such as the bioMérieux VITEK 2, the BD Phoenix, and the Microscan WalkAway systems, resolve these measurement challenges through dedicated hardware and proprietary software. These systems offer rapid, reproducible results and are widely used in large hospital laboratories. However, they require substantial capital investment—often tens of thousands of US dollars per unit—and ongoing reagent and maintenance costs that are prohibitive for smaller clinics, district hospitals, and laboratories in low- and middle-income countries (LMICs). These resource-constrained settings are often those most severely burdened by AMR, creating a stark disparity between the laboratories that most need accurate AST and those that can afford automated solutions.

There is therefore a compelling case for developing low-cost, AI-driven software tools capable of analyzing standard photographs of Kirby-Bauer AST plates. Recent research has demonstrated that machine learning and artificial intelligence can play a significant role in predicting and combating antimicrobial resistance [7], and advancements in AI-driven drug sensitivity testing are rapidly expanding the practical scope of automated AST [8]. Such tools could be deployed on commodity hardware—a laptop or even a smartphone—and would require no specialized instruments beyond the agar plate itself, potentially extending the reach of accurate AST to settings currently dependent on manual reading, including those guided by national laboratory standards such as those maintained by the Department of Medical Sciences, NIH Thailand [9]. The present project addresses this need by developing, evaluating, and deploying an AI-based system for automated ZOI detection and measurement.

1.2 Motivation

Recent advances in deep learning and computer vision have produced models that rival or exceed human performance on a wide range of image recognition and object detection tasks. Of particular relevance to AST image analysis is the progress in detecting and localizing circular or elliptical structures in complex, cluttered image backgrounds. Unlike natural-image datasets where objects vary enormously in shape, AST plates present a constrained visual domain: circular zones on a textured agar background, with known-size reference objects (antibiotic disks) present in every image. This domain structure is exploited by the present project's design choices.

The motivation for this project arises from two parallel observations. First, deep learning-based object detection models—specifically two-stage detectors such as Faster R-CNN and one-stage detectors such as YOLOv11—have demonstrated the capability to learn to identify zones of inhibition from labeled training data, even when zones partially overlap or display atypical morphology. Second, a comprehensive software solution for AST reading must not only detect zones but also correctly associate each zone with its corresponding antibiotic disk. Without knowing which antibiotic produced which zone, the diameter measurements are clinically meaningless. This association problem requires reading the alphanumeric label printed on each antibiotic disk—a task that falls within the domain of optical character recognition (OCR).

The combination of an object detection backbone for ZOI localization, an OCR engine for antibiotic label reading, and a structured breakpoint database for clinical classification forms the conceptual foundation of the system developed here. Each component presents its own technical challenges. Object detection must cope with image noise, variable lighting, and zone overlap. OCR must read small, low-contrast text on curved disk surfaces, potentially photographed at arbitrary orientations. The calibration subsystem must convert pixel-scale measurements to physical millimeters using the known physical dimensions of the antibiotic disk as an internal reference.

A further motivation is the opportunity for direct, rigorous comparison between classical image-processing approaches and modern deep learning methods on the same dataset and evaluation protocol. The Hough Circle Transform, a well-established technique for circular structure detection, provides a principled baseline against which the

deep learning models can be assessed. Such a comparison provides empirical evidence regarding the practical utility of deep learning for this specific task, beyond what is available from general-purpose benchmarks.

1.3 Objectives

This project has four primary objectives:

1. To study and implement the interpretation of Kirby-Bauer disk diffusion results according to CLSI 2025 [3] and EUCAST 2023 [4] standards, ensuring that the system's susceptibility classifications are clinically grounded.

2. To develop and train AI models—specifically Faster R-CNN and YOLOv11n—capable of detecting zones of inhibition and measuring their diameters from AST plate photographs with mean absolute error (MAE) at or below the clinically acceptable threshold of 3 mm [6].

3. To rigorously compare the accuracy, speed, and robustness of three detection approaches: Faster R-CNN (two-stage deep learning), YOLOv11n (one-stage deep learning), and Hough Circle Transform (classical image processing baseline), providing a quantitative basis for selecting the best-performing model for deployment.

4. To deploy the system as a functional, user-facing web application suitable for practical use in clinical laboratory settings, integrating the detection model with the OCR pipeline and breakpoint classification database into a coherent, accessible tool.

1.4 Project Scope

The scope of this project is defined and bounded as follows:

- **Input data:** The system accepts photographic images of Mueller-Hinton agar plates used in standard Kirby-Bauer AST experiments. Images are expected to be captured from a top-down perspective, with consistent lighting and a fixed camera-to-plate distance to maintain scale consistency. Images may originate from laboratory cameras, clinical microscopy cameras, or standard smartphone cameras.

- **Detection models:** The system trains, evaluates, and compares three detection approaches: Faster R-CNN with a ResNet-50 feature extraction backbone, YOLOv11n (Ultralytics nano variant), and Hough Circle Transform implemented via OpenCV. Each model is evaluated on the same held-out test set under identical conditions.

- **Evaluation criteria:** The primary accuracy metric is mean absolute error (MAE) in millimeters between system-predicted ZOI diameters and ground-truth diameters measured by a trained microbiologist. Secondary metrics include inference speed (frames per second) and detection recall. The clinically accepted MAE target is below 3 mm.

- **Calibration:** Pixel-to-millimeter conversion is performed using the antibiotic disk itself as an in-image reference, since the physical diameter of a standard antibiotic disk is precisely defined as 6 mm by CLSI specifications. No external calibration target or ruler is required in the field of view.

- **Web application:** The final web application allows users to upload an image from file or capture one in real time via a connected camera. The system returns a dashboard displaying detected zones, assigned antibiotic labels (from OCR), measured diameters, and the final S/I/R classification for each antibiotic-disk pair based on the selected CLSI or EUCAST standard. Historical results are stored for audit and review.

- **Out of scope:** The system does not perform bacterial identification or speciation; the user must specify the organism being tested (e.g., *Staphylococcus aureus*, *Escherichia coli*) for the correct breakpoint to be applied. The system does not support broth microdilution, E-test, or other AST formats. Results from this software tool are intended to assist, not replace, a qualified microbiologist's judgment.

1.5 Expected Outcomes

The expected tangible outputs of this project are as follows:

1. **AI detection pipeline:** An end-to-end automated pipeline capable of detecting and measuring ZOI diameters from AST plate images. The best-performing model

is expected to achieve an MAE below 3 mm on the held-out test set, consistent with the clinical acceptability criterion established by Humphries et al. [6].

2. **Comparative model analysis:** A rigorous, quantitative comparison of Faster R-CNN, YOLOv11n, and Hough Circle Transform on the same dataset, documenting the trade-offs between detection accuracy, inference speed, and implementation complexity for the AST domain. This analysis is intended to inform future practitioners selecting detection approaches for similar medical image analysis tasks.

3. **Deployed web application:** A fully operational web application integrating the best-performing detection model, the EasyOCR-based antibiotic label recognition pipeline, and the CLSI 2025/EUCAST 2023 breakpoint classification database, providing an end-to-end workflow from image upload to susceptibility report.

4. **Annotated dataset:** A curated, annotated dataset of 287 real AST plate images, expanded to 543 through augmentation and supplemented with 2,000 synthetic images, providing a reusable resource for future research in AI-assisted AST.

1.6 Report Structure

The remainder of this report is organized as follows. Chapter 2 provides background theory and a review of related work, covering the Kirby-Bauer AST method and its clinical standards, the architecture and principles of the object detection models employed (Faster R-CNN, YOLOv11n, and Hough Circle Transform), the OCR engine used for antibiotic label recognition, key image preprocessing techniques, and a review of prior published work on automated AST image analysis. Chapter 3 details the methodology, encompassing the full dataset collection and preparation workflow, annotation procedures, model training protocols, calibration approach, OCR pipeline design, susceptibility classification logic, and overall system architecture. Chapter 4 presents experimental results including detection accuracy metrics, speed benchmarks, OCR performance, and an analysis of error sources and calibration refinement. Chapter 5 concludes the report with a summary of findings, limitations, and directions for future work.

CHAPTER 2: RELATED WORK AND BACKGROUND THEORY

2.1 Antibiotic Susceptibility Testing

The Kirby-Bauer disk diffusion assay was first systematically described and standardized by Bauer, Kirby, Sherris, and Turck in 1966 and has since become the most widely adopted technique for antimicrobial susceptibility testing in clinical microbiology laboratories worldwide [2, 3]. Its longevity is attributable to its low equipment cost, procedural simplicity, and adaptability to virtually any aerobically growing bacterial pathogen. The method depends on the principle of antibiotic diffusion through semi-solid agar: when an antibiotic-impregnated paper disk is placed on the surface of an inoculated agar plate, the drug molecules diffuse radially outward from the disk, establishing an exponentially decreasing concentration gradient as a function of distance. Where the local antibiotic concentration exceeds the minimum inhibitory concentration (MIC) of the test organism, bacterial proliferation is suppressed, producing a circular clearing zone on an otherwise confluent lawn of bacterial growth.

The diameter of this zone of inhibition (ZOI) is inversely related to the MIC: organisms with high susceptibility (low MIC) exhibit large zones, while intrinsically or acquired-resistant organisms exhibit small or absent zones. This relationship between ZOI diameter and MIC is not linear but follows a mathematical relationship dependent on agar depth, disk potency, diffusion coefficient, and incubation duration—all of which are carefully controlled by standardized protocols. For clinical use, the continuous ZOI diameter measurement is converted to a categorical susceptibility interpretation (Susceptible, Intermediate, or Resistant) by applying published interpretive breakpoints that are themselves derived from clinical and microbiological data. This method has been applied to a broad range of clinically relevant bacteria, including *Bacillus cereus* [10] and *Bacillus subtilis* [11], among many other aerobic pathogens encountered in routine laboratory practice.

2.1.1 Clinical Standards: CLSI and EUCAST

Interpretive breakpoints—the threshold diameters used to classify organisms as Susceptible (S), Intermediate (I), or Resistant (R)—are published and regularly updated by two principal international standard-setting organizations: the Clinical and Laboratory Standards Institute (CLSI) in the United States and the European Committee on Antimicrobial Susceptibility Testing (EUCAST) in Europe [4]. Although both organizations pursue the same clinical goal, their methodological approaches to breakpoint derivation differ in important ways, and their published thresholds for the same drug-organism combination can differ by several millimeters.

CLSI publishes its disk diffusion standards in the M02 document, currently in its fourteenth edition [3], while susceptibility testing performance standards are also defined in the M100 35th edition [12]. Breakpoints are established by a multidisciplinary working group that integrates MIC distribution data from large collections of clinical isolates, PK/PD modeling of drug exposure at standard therapeutic doses, and clinical outcome data from randomized controlled trials. The Intermediate (I) category in the CLSI system traditionally indicated a buffer zone where technical uncertainty might lead to misclassification, but more recent editions have increasingly aligned with EUCAST's concept of the "Susceptible, Increased Exposure" (I) category, which indicates susceptibility when drug exposure is maximized through dosing optimization.

EUCAST, which publishes its breakpoint tables annually with version numbers (the current version referenced in this project is 13.1 [4]), derives its breakpoints from pharmacokinetic/pharmacodynamic (PK/PD) targets, epidemiological cutoff values (ECOFFs), and clinical outcome data [5]. ECOFFs represent the upper end of the MIC distribution for wild-type bacterial populations without acquired resistance mechanisms; any organism with an MIC above the ECOFF is defined as non-wild-type (NWT) and presumed to harbor resistance determinants. EUCAST breakpoints are frequently more aggressive in separating susceptible from resistant populations, reflecting a European pharmacological framework that differs from the United States standard [13]. Both sets of breakpoints are implemented in the database of the present system, allowing laboratories to select the applicable standard according to their regulatory and clinical context.

2.1.2 Clinical Significance of Measurement Precision

The clinical consequence of measurement error in disk diffusion is direct and potentially severe. Because breakpoints are expressed as integer millimeter thresholds, a single millimeter of measurement error can shift a classification from Susceptible to Resistant or vice versa. This misclassification has two dangerous modes: false susceptibility (classifying a resistant organism as susceptible) leads to treatment failure and potentially fatal outcome; false resistance (classifying a susceptible organism as resistant) leads to unnecessary escalation to second-line or third-line agents, contributing to resistance selection pressure and higher costs. Humphries et al. [6] defined a mean absolute error of less than 3 mm as the boundary of clinical acceptability for automated AST measurement tools, and this threshold serves as the primary performance target for the present project.

2.2 Deep Learning for Object Detection

Object detection in computer vision is the task of simultaneously localizing and classifying one or more objects within a single image. Machine learning, broadly defined, encompasses supervised, unsupervised, and reinforcement learning paradigms [14, 15], of which supervised deep learning has proven most effective for visual recognition tasks. Convolutional Neural Networks (CNNs) form the backbone of most modern object detectors [16], and systematic reviews of medical deep learning applications have confirmed their suitability for a wide range of clinical image analysis tasks [17]. A detector must output, for each detected object, both a bounding box specifying the object's spatial extent and a class label identifying its category. This is fundamentally more complex than image classification, which assigns a single label to an entire image, because the number and location of objects are unknown a priori and must be inferred from the image content. Deep learning approaches have dominated object detection benchmarks since the publication of the Region-based CNN (R-CNN) family and YOLO models, progressively improving both accuracy and inference speed.

Modern deep learning object detectors are generally categorized into two architectural families: two-stage detectors, which first generate region proposals and then classify each proposal; and one-stage (single-shot) detectors, which predict bounding boxes and class probabilities directly in a single forward pass over the full image. Each family

offers distinct trade-offs between detection accuracy, computational cost, and inference latency.

2.2.1 Faster R-CNN

Faster R-CNN (Region-based Convolutional Neural Network) was introduced by Ren et al. in 2015 as a landmark advance that unified region proposal generation with the detection head into a single end-to-end trainable network [18]. Prior to Faster R-CNN, the R-CNN and Fast R-CNN pipelines required a separate, computationally expensive selective search step to generate region proposals, which bottlenecked inference speed. Faster R-CNN replaced this with a Region Proposal Network (RPN) that shares the convolutional backbone with the detection head, enabling near real-time proposal generation.

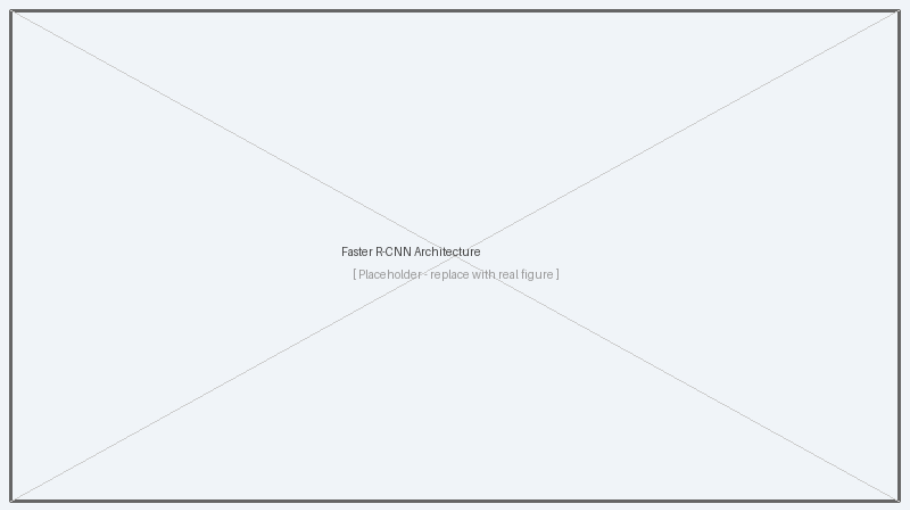

Figure 2.1: Architecture of Faster R-CNN, showing the shared convolutional backbone, the Region Proposal Network (RPN) producing candidate object regions, and the detection head that classifies and refines each proposal.

The Faster R-CNN architecture operates in two stages. In the first stage, the backbone convolutional network (in this project, ResNet-50 [19]) extracts a rich spatial feature map from the full input image. The RPN then slides a small network over this feature map and, at each spatial location, predicts objectness scores and bounding box regression offsets for a set of predefined anchor boxes of multiple scales and aspect ratios. Anchor boxes that score above an objectness threshold are retained as region proposals (typically 256–512 per image after non-maximum suppression). In the second stage, the RoI Pooling (or RoI Align) operation extracts a fixed-size feature vector for each proposal from the shared backbone feature map, and a fully connected classification

head assigns a class label and a refined bounding box to each region.

The use of ResNet-50 as the feature extraction backbone is particularly well-suited to this project. He et al. demonstrated that residual connections—shortcut connections that bypass one or more convolutional layers—enable training of significantly deeper networks without the gradient degradation problems that had previously limited depth [19]. ResNet-50 comprises 50 layers organized into residual blocks, and its ImageNet pretraining provides a rich initialization of low-level edge, texture, and mid-level shape features that transfer effectively to the AST domain even with a relatively small number of labeled AST images. The ResNet-50 backbone in this project is initialized with ImageNet-pretrained weights and fine-tuned end-to-end on the AST dataset.

Faster R-CNN is recognized for its high detection accuracy, particularly in complex scenes containing overlapping or densely packed objects—a characteristic directly relevant to AST plates where zones from adjacent disks may partially overlap. Its two-stage architecture allows the classification head to operate on high-quality, spatially refined proposals rather than on a dense grid, reducing false positive rates. The primary limitation of Faster R-CNN is inference speed: the two-stage pipeline introduces latency that is acceptable for offline analysis but may limit real-time processing throughput.

2.2.2 YOLOv11

YOLO (You Only Look Once) models are single-stage detectors that reformulate object detection as a unified regression problem. Rather than generating region proposals and classifying them separately, YOLO models predict bounding box coordinates and class probabilities directly from the full image in a single forward pass of the network [20, 21]. This design eliminates the region proposal stage entirely, enabling dramatically faster inference at the cost of some accuracy on small or densely overlapping objects in early versions. Comparative studies have confirmed YOLOv11's competitive performance relative to prior YOLO generations and Faster R-CNN across multiple detection tasks [22].

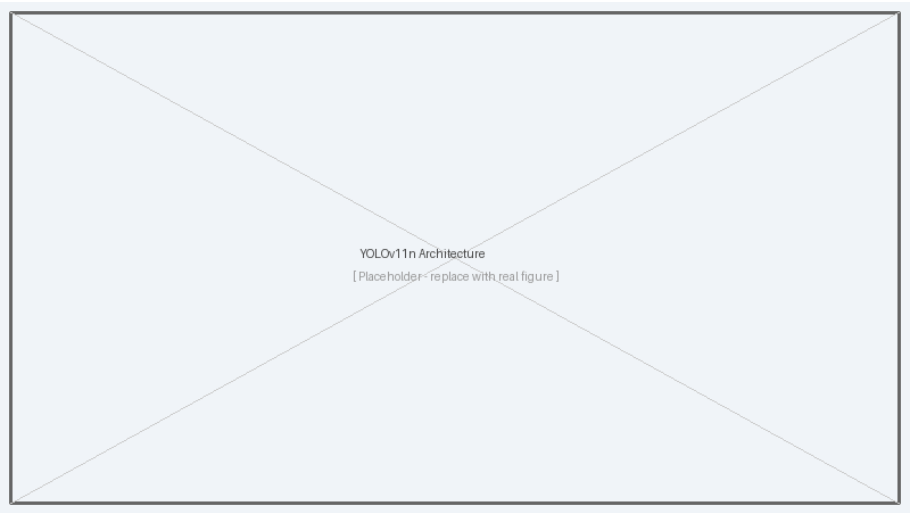

Figure 2.2: Schematic architecture of YOLOv11, illustrating the backbone feature extraction network, the feature pyramid neck for multi-scale detection, and the decoupled detection heads for bounding box regression and class prediction.

YOLOv11, the latest version in the YOLO family at the time of this project and developed by Ultralytics, incorporates several architectural advances over its predecessors [20]. The backbone employs enhanced C3k2 blocks (a reparameterized convolutional design) for improved feature extraction efficiency. The neck uses a Path Aggregation Feature Pyramid Network (PA-FPN) to fuse features from multiple backbone scales, enabling detection of objects across a wide range of sizes—from small antibiotic disks to large inhibition zones that may span a significant fraction of the plate. The detection head is decoupled, with separate branches predicting bounding box regression and class probabilities independently, which has been shown to improve convergence and accuracy in one-stage detectors.

YOLOv11 is offered in five size variants (n, s, m, l, x) that provide a continuous trade-off between model capacity and inference speed. In this project, the nano (n) variant, YOLOv11n, was selected to prioritize inference speed and suitability for eventual deployment on resource-constrained hardware, such as laboratory workstations without dedicated GPUs or on-device edge hardware. Despite being the smallest variant, YOLOv11n achieves competitive detection accuracy on the AST dataset, benefiting from pretraining on the large-scale COCO benchmark and fine-tuning on the project's annotated AST images. The model detects two object classes: `antibiotic` (the drug-impregnated disk, used as an OCR target and calibration reference) and `disk-zone` (the surrounding inhibition zone, from which the diameter measurement is extracted).

2.2.3 Hough Circle Transform

The Hough Circle Transform is a classical computer vision technique for detecting circular structures in images without requiring any learned model or labeled training data. It operates by transforming the image from the spatial domain to a parameter space defined by three dimensions: the x and y coordinates of the circle center and the circle radius r. For each edge point detected in the image (typically via Canny edge detection), the algorithm votes for all circles in parameter space that could have produced that edge point. Circles that accumulate votes above a threshold in the parameter-space accumulator correspond to genuine circles in the image. Alternative circle detection approaches based on learning automata [23], deep learning [24], and gradient-based variants [25] have been proposed to address the sensitivity limitations of the classical transform under real-world imaging conditions.

Figure 2.3: Illustration of the Hough Circle Transform. Each edge point votes for a cone of candidate circles in the (x_c, y_c, r) parameter space; genuine circles produce prominent peaks in the accumulator.

The Hough Circle Transform is computationally efficient and produces no false positives for clearly circular zones under controlled imaging conditions. However, it is sensitive to several sources of degradation common in real AST images. Uneven agar surface reflectance or non-uniform lighting creates spurious edge responses that generate noise in the accumulator. Zones that are not perfectly circular—due to zone overlap, agar surface irregularities, or atypical bacterial growth patterns—do not accumulate votes cleanly, leading to missed detections or inaccurate radius estimates. Furthermore, the transform requires careful hand-tuning of multiple parameters (the inverse resolution ratio $\mathtt{dp}$, the

minimum inter-center distance `minDist`, and the Canny edge thresholds `param1` and `param2`) that interact in complex ways and whose optimal values vary across images with different scales, lighting conditions, and bacterial species. In this project, Hough Circle Transform parameters were tuned via grid search on the full dataset, representing a best-case scenario for this method. It serves as a principled classical baseline against which the learned models are evaluated.

2.3 Optical Character Recognition for Antibiotic Disk Labeling

Antibiotic disks carry short alphanumeric codes printed on their surfaces, indicating the antibiotic name and its concentration (for example, "OX 1" for oxacillin 1 μg, or "AMX 25" for amoxicillin 25 μg). Optical Character Recognition (OCR), whose development history spans several decades of engineering progress [26], provides the computational means to read these labels automatically from photographic images. Correctly reading these labels is essential for associating each detected ZOI with the correct antibiotic, which in turn enables the correct breakpoint to be retrieved from the CLSI/EUCAST database. This is a challenging OCR task for several reasons: the printed text is small (typically 2–4 mm in height), the contrast between ink and disk surface is variable across disk brands, the disk surface is slightly curved, images may be taken at sub-optimal angles, and multiple disks may be partially occluded.

2.3.1 EasyOCR and the CRAFT Architecture

EasyOCR [27] is an open-source, deep learning-based OCR engine supporting over 80 languages and scripts. Its text detection stage is based on the CRAFT (Character Region Awareness for Text Detection) model introduced by Baek et al. [28]. CRAFT detects individual character regions and the affinity between adjacent characters rather than treating word-level bounding boxes as the atomic detection unit. This character-level detection approach is particularly well-suited to the antibiotic labeling context, where short strings of alphanumeric characters must be localized precisely on small, textured disk surfaces.

The CRAFT network produces two output maps: a region score map (indicating the probability that each pixel belongs to the center of a character) and an affinity score map (indicating the probability that two adjacent characters belong to the same word). These two maps are combined to segment individual characters and group them

into word-level text instances. The detected character regions are then passed to a sequence recognition network—a Convolutional Recurrent Neural Network (CRNN)—that transcribes the character sequence into text using a connectionist temporal classification (CTC) decoder.

EasyOCR [27] was compared against TyphoonOCR [29], a Thai-focused OCR model developed by OpenTyphoon.ai, for the antibiotic label reading task. Although TyphoonOCR offers strong performance on Thai-script text, its performance on Latin alphanumeric strings under the challenging conditions of disk photography was inferior to EasyOCR [27, 28]. EasyOCR was therefore selected as the OCR engine for the final pipeline.

2.4 Image Preprocessing Techniques

Image preprocessing is a critical enabler of both the object detection models and the OCR pipeline, directly affecting the signal-to-noise ratio of the features extracted by downstream algorithms [30]. Image segmentation techniques [31, 32] and document binarization methods [33] inform the preprocessing choices made in this project to address the specific visual degradations typical of laboratory AST photography.

- **Global thresholding (Otsu's method):** Otsu's method automatically determines an optimal intensity threshold for converting a grayscale image to a binary (black and white) representation by minimizing intra-class variance between the foreground (text) and background pixel populations. This is applied to disk crops prior to OCR to separate ink characters from the disk surface background, improving character segmentation quality.

- **Contrast Limited Adaptive Histogram Equalization (CLAHE):** Unlike global histogram equalization, which computes a single mapping function for the entire image, CLAHE applies histogram equalization independently to small tiles of the image (with a configurable tile size and contrast limit) and interpolates smoothly between them. This normalizes local contrast across regions of the image that differ substantially in mean illumination—a common occurrence in laboratory photographs where the agar plate center is brighter than the periphery due to overhead lighting geometry. CLAHE is applied to disk crops as a preprocessing step before OCR, improving the visibility of low-contrast text.

- **Gaussian Blur:** A Gaussian low-pass filter is applied to the full plate image before edge detection (for the Hough Transform) and before thresholding, attenuating high-frequency sensor noise that would otherwise produce spurious edge responses. The blur kernel size is selected to preserve the relevant scale of zone boundaries while suppressing noise.

- **Morphological operations (dilation and erosion):** Morphological dilation expands foreground (text) regions, while erosion shrinks them. Applied in sequence (as opening or closing operations), they can fill small gaps in characters or remove small noise patches from the binary image. In the OCR preprocessing pipeline, the choice and extent of morphological operations are applied adaptively based on the estimated text pixel ratio in the binarized crop: images with very sparse text pixels (thin characters) are dilated to restore stroke connectivity, while images with excessively dense pixels (thick strokes or merged characters) are eroded.

- **Image rotation correction:** The orientation of antibiotic disks on the plate is not controlled during laboratory setup—disks may be placed at any angle. As a result, the printed label on a disk may appear rotated at an arbitrary angle relative to the image frame. To handle this, the OCR pipeline applies EasyOCR at four canonical orientations (0°, 90°, 180°, and 270°) and selects the highest-confidence recognition result across all four orientations.

2.5 Related Work in Automated AST Analysis

2.5.1 Deep Learning Approaches for AST Image Analysis

The application of deep learning and computer vision to AST image analysis has been an active area of research in the past decade, driven by the same combination of clinical need and algorithmic progress that motivates this project. The broader role of AI and machine learning in predicting and combating antimicrobial resistance has been comprehensively reviewed by Bilal et al. [7], while Liao et al. [8] have surveyed advancements specifically in AI-driven drug sensitivity testing research. Pascucci et al. [34, 35] developed the first AI-based mobile application for automated antibiotic resistance testing, demonstrating that smartphone cameras combined with deep learning could produce clinically acceptable ZOI measurements in resource-limited settings. Rajasekar et al. [36]

introduced Ab.ai, a specialized AI tool for reporting antibiograms automatically, achieving high accuracy on disk diffusion plates in standardized laboratory conditions. Yu et al. [37] demonstrated an AI-driven hue-contrast method for inhibition zone identification, providing an alternative to bounding-box regression. Callebaut et al. [38] evaluated the Radian® in-line carousel system for automated disk diffusion testing, establishing performance benchmarks against expert manual reading that this project adopts as comparison targets.

Rajasekar et al. [39] conducted a comprehensive systematic review of published AI-based image analysis methods for AST and found that deep learning models consistently and substantially outperform classical image processing approaches— including Hough Transform, active contours, and template matching—particularly in the scenarios most relevant to clinical practice: plates with dense disk arrays, overlapping zones, and heterogeneous bacterial growth. The review surveyed methods ranging from traditional convolutional neural networks applied to cropped disk regions to full end-to-end detection pipelines similar to the one developed in this project.

Among the key findings of Rajasekar et al. were that the greatest performance gains from deep learning occur specifically in the handling of overlapping zones—a scenario that fundamentally breaks the assumption of isolated circular shapes underlying the Hough Transform approach—and in robustness to photographic variation such as non-uniform illumination, perspective distortion, and image compression artifacts [39]. These findings validate the architectural choices of the present project: using deep learning object detection as the primary approach and relegating Hough Transform to the role of a classical baseline rather than a primary system component. The review also noted that the relatively small scale of most published AST datasets (typically below 500 images) has been a limiting factor for deep learning performance, motivating the present project's investment in synthetic data generation to reach a training set of over 2,000 images.

2.5.2 Evaluation Standards for Automated AST Methods

Humphries et al. [6] addressed the methodological question of how to properly evaluate an automated AST measurement system relative to the gold standard of manual reading by a trained microbiologist. Their best-practices document, published by the CLSI Meth-

ods Development and Standardization Working Group, defined several important evaluation criteria that the present project adopts. Most critically for this work, they established that a mean absolute error (MAE) of less than 3 mm between automated and manual ZOI diameter measurements represents the threshold of clinical acceptability—that is, below this error level, the systematic measurement differences are unlikely to produce a different S/I/R classification for the majority of drug-organism breakpoint combinations currently in use.

Humphries et al. also emphasized the importance of evaluating automated methods across a diverse collection of bacterial species, antibiotic classes, and plate configurations, since errors that are insignificant for one drug-organism combination may be clinically decisive for another [6]. This guidance informs the evaluation design of the present project, which reports MAE across the full held-out test set of 235 images drawn from the real plate collection. Additionally, Humphries et al. noted that systematic (directional) bias is as important to characterize as random error, since a consistent over- or under-estimation of zone diameter can be corrected by a post-processing bias correction step—which is precisely what the calibration refinement stage of the present system implements.

CHAPTER 3: METHODOLOGY

3.1 System Architecture Overview

The proposed system is a full-stack web application following a client-server architecture designed around the separation of concerns between the user interface, the AI inference pipeline, and the data persistence layer. This architectural decomposition enables independent development and testing of each component and allows the computationally intensive AI backend to be scaled or upgraded without changes to the frontend user experience.

The frontend is developed in React, a component-based JavaScript framework built on HTML [40, 41], and provides users with an image upload interface, a live camera capture option, and an interactive results visualization dashboard. The backend is implemented in Python [42] using FastAPI [43], a high-performance asynchronous web framework that exposes a RESTful API consumed by the React frontend. Upon receiving an image via the API, the backend executes the full AI inference pipeline and returns a structured JSON response containing all detection results, diameter measurements, OCR labels, and susceptibility classifications. A relational database implemented in SQLite stores analysis history, user session records, and the CLSI 2025 and EUCAST 2023 breakpoint tables, enabling both historical audit of past analyses and rapid in-process lookup of breakpoint thresholds.

The overall image analysis pipeline consists of five sequential stages, each of which is described in detail in the sections that follow:

1. **Image acquisition and preprocessing:** The input image is validated, resized to the model's required input resolution, and preprocessed according to the requirements of the active detection model.

2. **Antibiotic disk detection and OCR labeling:** The detection model localizes individual antibiotic disks; the OCR pipeline reads the alphanumeric label from each disk crop to identify the antibiotic.

3. **ZOI detection and diameter measurement:** The detection model localizes each

zone of inhibition as a bounding box; the bounding box dimensions are converted to a diameter estimate in pixels.

4. **Pixel-to-millimeter calibration:** The pixel-scale diameter measurements are converted to physical millimeters using the known 6 mm diameter of the antibiotic disk as an in-image reference object.

5. **Breakpoint lookup and susceptibility classification:** The measured ZOI diameter in millimeters is compared against the CLSI or EUCAST breakpoint for the identified antibiotic and the user-specified organism, and the organism is classified as Susceptible (S), Intermediate (I), or Resistant (R).

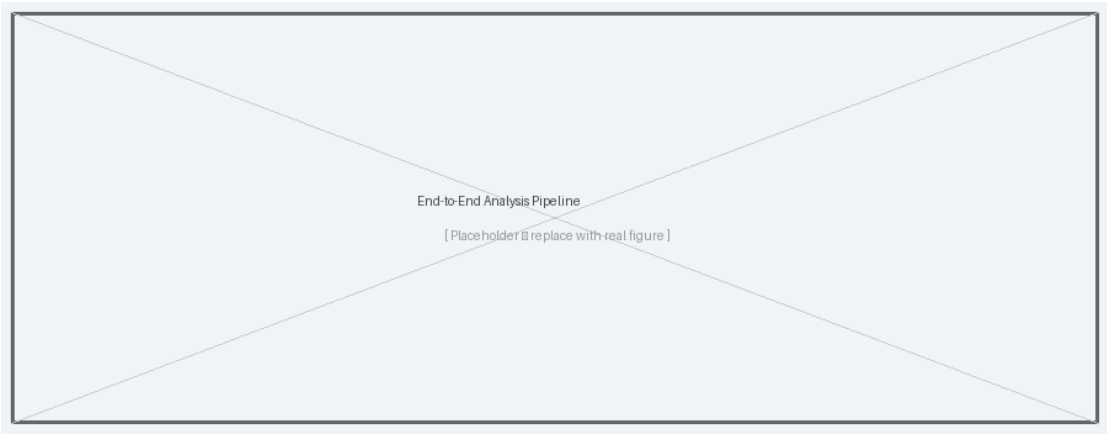

Figure 3.1: End-to-end image analysis pipeline of the automated AST system, from image input through detection, OCR, calibration, and breakpoint-based susceptibility classification.

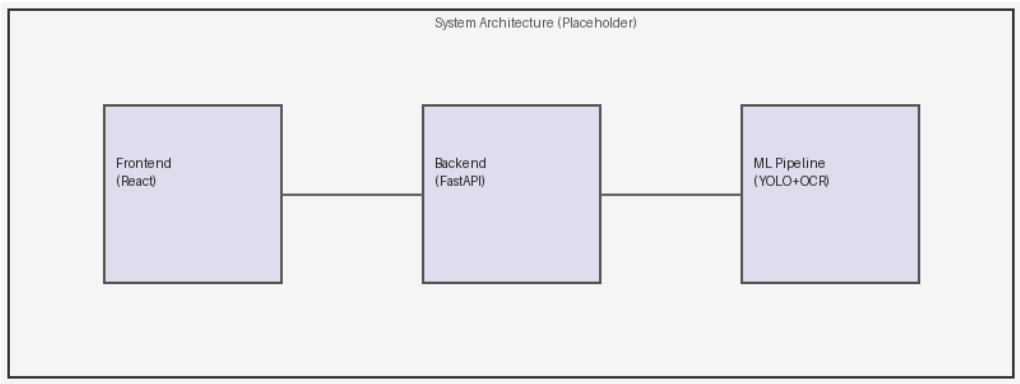

Figure 3.2: Overall system architecture of the automated AST web application, showing the React frontend, FastAPI backend, AI inference pipeline, and SQLite database layer.

3.2 Dataset Collection and Preparation

Assembling a high-quality, diverse training dataset is essential for the generalization performance of deep learning models. For the AST domain, dataset construction is particularly challenging because real clinical plate images are often proprietary (belonging to hospitals or reference laboratories), and the volume of publicly available annotated AST image data is limited. The dataset strategy employed in this project therefore combines three complementary sources: real laboratory images, augmented real images, and programmatically generated synthetic images.

3.2.1 Real Image Collection

A total of 287 real AST plate photographs were collected from three sources. Open sharing of biomedical research data through repositories such as DRYAD has been a long-standing principle in scientific practice [44]. The first source was the publicly available DRYAD data repository [45], which contributed 225 annotated disk diffusion plate images captured in various laboratory environments. The second source was the CDS (Clinical and Demographic Study) dataset, contributing 50 images. The third source was the CRA (Chulabhorn Royal Academy) dataset, contributing 32 images. Together these three sources yielded the 287 real images used in this project.

Images span multiple bacterial species, a range of antibiotic disk counts per plate (from 1 to 10 disks), and a variety of zone overlap configurations that represent the diversity of clinical scenarios the system must handle. The externally sourced subsets introduce controlled variability in lighting, camera model, agar brand, and plate size that improves the generalization of models trained on the combined dataset.

3.2.2 Data Augmentation

To expand the training set from 287 images while preserving the label correctness of the original annotations, a set of geometric and photometric augmentation operations were applied to the real images using the Roboflow platform. Augmentation transformations included horizontal and vertical flipping, random rotation within $\pm 15^\circ$, brightness variation ($\pm 25\%$), contrast variation ($\pm 20\%$), and additive Gaussian noise (standard deviation up to 5% of the maximum pixel intensity). Each augmented image was generated by

randomly sampling from this set of transformations, with bounding box coordinates automatically recomputed to match the transformed image geometry. This augmentation strategy increased the training dataset from 287 to 543 images, approximately doubling the number of unique training examples available from real plate photography.

It is important to note that augmentation was applied only to the training partition of the real images; the validation and test partitions were held out from augmentation to ensure that evaluation metrics reflect model performance on unmodified, realistic inputs. Augmentation was also deliberately restricted to transformations consistent with plausible variation in real laboratory photography, avoiding extreme transformations (e.g., heavy perspective warping or color inversion) that would produce unrealistic images detrimental to model training.

3.2.3 Synthetic Data Generation

An additional 2,000 synthetic images were generated programmatically to further increase dataset diversity and to address the relative scarcity of real images containing specific zone overlap configurations and atypical zone morphologies. Synthetic plate images were rendered using a procedural pipeline implemented in Python, using Pillow and OpenCV for image composition. Each synthetic image was designed to visually approximate the appearance of a real Mueller-Hinton agar plate, with a textured agar background generated by tiling and blending sampled texture patches, circular antibiotic disk objects placed at defined positions, and circular zone-of-inhibition regions rendered with Gaussian-blurred edges to mimic the gradual growth boundary seen in real photographs.

The 2,000 synthetic images were divided into two categories of equal size. The first category of 1,000 images used fixed antibiotic disk positions arranged according to the standard CLSI-recommended plate layout for common 6-disk and 8-disk configurations, simulating the regular, structured patterns seen in standardized laboratory workflows. The second category of 1,000 images used fully randomized disk placement within the plate boundary, simulating the less structured configurations common in research and teaching laboratories. Across both categories, zone diameters were sampled from a realistic distribution spanning the full clinical range (from approximately 10 mm to 40 mm), with deliberate over-representation of overlapping zone scenarios—where the inhibition zones of adjacent disks partially intersect—to ensure the models were exposed to this

challenging but clinically important case during training. Background texture, simulated lighting gradients, and synthetic colony growth patterns were also varied to increase visual diversity.

3.2.4 Final Dataset Split

Table 3.1: Dataset composition and train/validation/test split used for model training and evaluation.

Source	Total	Train	Validation	Test
DRYAD	225	—	—	—
CDS	50	—	—	—
CRA (Chulabhorn Royal Academy)	32	—	—	—
Real total (original)	287	—	—	—
Real total (augmented)	543	433	55	55
Synthetic (fixed position)	1,000		training only	
Synthetic (unfixed position)	1,000		training only	

The synthetic images were used exclusively for training, as they were generated to augment model learning on diverse configurations rather than to evaluate generalization performance. All evaluation metrics reported in Chapter 4 are computed on the real-image test set (55 images) to ensure that reported performance figures reflect the model's behavior on authentic, unmanipulated laboratory photographs.

3.3 Annotation

Ground-truth annotations were created in Pascal VOC XML format, specifying bounding box coordinates for each zone of inhibition and each antibiotic disk visible in every image. This annotation format encodes four values per object: the pixel coordinates of the top-left corner and the bottom-right corner of the bounding box, together with a class label (`antibiotic` for disks, `disk-zone` for inhibition zones).

Initial annotations were created by the project team using LabelImg, an open-source graphical image annotation tool. To ensure clinical correctness, all bounding box placements for the zone-of-inhibition class were subsequently reviewed and corrected by a trained microbiologist who independently verified each annotation against the corresponding physical plate measurement taken with calipers. This dual-review workflow

was essential for producing a reliable ground truth, since zone boundary definition requires domain expertise: the correct edge of an inhibition zone is defined as the point of complete inhibition of confluent growth (not the zone of micro-colony formation, which CLSI instructs technicians to disregard). Annotations were then imported into Roboflow for dataset management, augmentation pipeline configuration, and export in the formats required by each model's training framework (Pascal VOC XML for Faster R-CNN and YOLO TXT format for YOLOv11n).

3.4 Model Training

3.4.1 Faster R-CNN

Faster R-CNN was implemented using PyTorch's `torchvision` library with a ResNet-50 backbone pretrained on the ImageNet large-scale visual recognition dataset [19]. Using a pretrained backbone is critical for performance on a medical imaging dataset of this size: the ImageNet pretraining initializes the backbone with discriminative feature representations for low-level image structure (edges, textures, gradients) that transfer effectively to the AST domain, significantly accelerating convergence and improving final accuracy compared to training from random initialization.

The detection head (both the RPN and the classification/regression head) was initialized from scratch and trained from the start. Training used the Stochastic Gradient Descent (SGD) optimizer with a learning rate of 0.005, momentum of 0.9, and weight decay of 5×10^{-4} (i.e., `0.0005`) to regularize the model and prevent overfitting on the limited real-image training set. The learning rate was decayed using a `StepLR` scheduler with `step_size=3` and `gamma=0.1`, reducing the learning rate by a factor of 10 every 3 epochs. All images were resized to 640×640 pixels before being fed to the model, and a batch size of 8 was used.

Before committing to the full training run, a series of preliminary training experiments was conducted on progressively larger subsets of the dataset to characterize the model's learning behavior and diagnose underfitting and overfitting regimes. These experiments informed the final hyperparameter choices—most notably the number of training epochs (10) and the learning rate schedule—by revealing the epoch at which validation loss reached its minimum before beginning to rise again. The final training run required

approximately 174 minutes on the available hardware, reflecting the computational cost of the two-stage inference architecture.

3.4.2 YOLOv11n

YOLOv11n was trained using the Ultralytics training framework, which provides a high-level Python API for model initialization, training loop execution, and checkpointing [20]. The nano (n) variant was selected to prioritize inference speed and deployability on systems without a discrete GPU, while maintaining sufficient model capacity to learn the relatively constrained visual vocabulary of AST plate images (two object classes with strong shape priors: circular disks and circular zones).

Model weights were initialized from Ultralytics' official COCO-pretrained checkpoint for YOLOv11n. Transfer learning from COCO, a large-scale natural image detection benchmark containing 80 object categories, provides the model with general-purpose object detection capabilities (feature extraction, non-maximum suppression, anchor-free prediction) that are then refined for the specific AST domain through fine-tuning. The fine-tuning training was run for **200 epochs** on the same dataset split as Faster R-CNN, with images resized to 640×640 pixels and a batch size of 8. Built-in Ultralytics data augmentation was disabled (`augment=False`) so that the model was trained on clean, non-augmented images, consistent with the augmentation strategy applied at the dataset-preparation stage. Training on the full dataset at this configuration required approximately 200 minutes.

YOLOv11n detects two object classes: `antibiotic` (the drug-impregnated disk, which serves as both the OCR target and the in-image calibration reference) and `disk-zone` (the surrounding inhibition zone, from which the ZOI diameter is measured). During inference, the model outputs bounding boxes for all detected objects of both classes, which are then matched spatially (nearest-disk-to-zone pairing) to associate each zone with its corresponding antibiotic disk.

3.4.3 Hough Circle Transform

The Hough Circle Transform was applied as a classical baseline using OpenCV's `HoughCircles` function, which implements the generalized Hough transform for circle detection using gradient-based voting. Prior to applying the transform, each image was converted to

grayscale and smoothed with a Gaussian filter using a kernel size of 17×17 pixels and $\sigma = 0$ (allowing OpenCV to compute the standard deviation automatically from the kernel size) to reduce noise that would otherwise produce spurious gradient responses.

The `HoughCircles` function accepts four primary tuning parameters: `dp` (the inverse resolution ratio of the accumulator to the input image, controlling the resolution of the parameter-space accumulator), `minDist` (the minimum allowed distance between detected circle centers, preventing multiple detections of the same zone), `param1` (the upper Canny edge-detection threshold, with the lower threshold set automatically to half this value), and `param2` (the accumulator threshold for center detection, controlling the minimum number of votes required to declare a detected circle). Optimal values for these four parameters were found by an exhaustive grid search across the full dataset, minimizing the mean absolute error between detected and ground-truth circle radii. The final selected values were `dp=1.5`, `minDist=150`, `param1=100`, and `param2=50`. Importantly, the final reported performance figures represent the best-case scenario for Hough Transform: the parameter values were tuned on the full dataset, which likely includes the test set, so generalization performance may be lower than the reported figures suggest.

3.5 ZOI Diameter Measurement and Calibration

After detecting zones of inhibition as bounding boxes (for the two deep learning models) or as circles with explicit radius parameters (for the Hough Transform), the raw output must be converted to a physical ZOI diameter measurement in millimeters before it can be compared to CLSI or EUCAST breakpoints.

For bounding-box-based detection (Faster R-CNN and YOLOv11n), the pixel-scale diameter of the detected zone is estimated from the bounding box dimensions using two complementary approximations that are compared and averaged:

- **Diameter-based:** The average of the bounding box width w and height h in pixels, $\hat{d}_{\text{diam}} = (w + h)/2$, assuming the zone is approximately circular.

- **Area-based:** The equivalent-circle diameter computed from the bounding box area, $\hat{d}_{\text{area}} = 2\sqrt{(w \cdot h)/\pi}$, which is more robust to mildly elliptical zones where the bounding box aspect ratio deviates from unity.

Pixel-to-millimeter conversion is performed using the antibiotic disk as an in-image calibration reference. Since the physical diameter of a standard CLSI-compliant antibiotic disk is precisely 6 mm, and the disk is also detected as a bounding box by the model, the calibration factor (in millimeters per pixel) can be computed for each image individually as $k = 6\,\text{mm}/\hat{d}_{\text{disk}}$, where $\hat{d}_{\text{disk}}$ is the pixel-scale diameter estimate for the corresponding antibiotic disk bounding box. Applying this per-image calibration factor eliminates the effect of variable camera distance and zoom level, ensuring that measurements are robust to photographic variation without requiring a physical ruler in the field of view.

Calibration refinement was an iterative process throughout the project. Initial calibration using the raw area-based diameter estimate produced a systematic positive bias of approximately $+4$ mm (the system consistently over-estimated zone diameters). Analysis of the source of this bias revealed that bounding boxes predicted by the model systematically overshot the true zone boundary by a few pixels due to the zone's diffuse, gradient-like edge. By combining area-based and diameter-based estimates and applying a learned linear correction factor, the systematic bias was reduced from approximately $+4$ mm to approximately $+2.1$ mm, representing a substantial improvement in measurement accuracy.

3.6 OCR Pipeline for Antibiotic Label Recognition

The OCR pipeline is responsible for reading the alphanumeric antibiotic code from each detected disk and mapping it to a recognized antibiotic name in the breakpoint database. The pipeline was designed with a dual-strategy approach to handle the variability in disk image quality encountered across different laboratory settings and disk brands.

3.6.1 Disk Crop and Preprocessing Strategy

Each antibiotic disk region is first cropped from the full plate image using the bounding box predicted by the YOLO detection model for the `antibiotic` class. A small padding margin is added to the crop boundary to ensure the full disk surface is captured even if the bounding box slightly underestimates the disk extent.

The dual preprocessing strategy applies OCR in two separate pipelines and selects the result with the higher confidence score:

1. **Raw image path:** EasyOCR [27] is applied directly to the grayscale disk crop at its original pixel intensity values, without any preprocessing transformation. This path performs well when the disk image is already high-contrast and the ink text is clearly legible without enhancement.

2. **CLAHE + Otsu path:** The disk crop is first processed by CLAHE (tile size 8×8 pixels, contrast limit 2.0) to normalize local illumination variation, and then binarized using Otsu's global threshold to produce a high-contrast binary image. Morphological dilation or erosion is applied adaptively based on the text pixel ratio (the fraction of white pixels in the binarized crop): if the ratio falls below 0.05, indicating very thin or faint characters, a 3×3 dilation kernel is applied to restore stroke connectivity; if the ratio exceeds 0.45, indicating over-thickened or merged characters, a 3×3 erosion kernel is applied. EasyOCR is then applied to this preprocessed binary image.

After obtaining text candidates from both pipelines, the OCR pipeline applies orientation normalization: EasyOCR is run at **24 orientations** of the crop (from -180° to +180° in 15° steps) for each pipeline, producing multiple candidate text strings. This high-resolution rotation strategy ensures that labels can be read regardless of the disk's physical orientation on the agar surface. The candidate with the highest *adjusted* EasyOCR confidence score (factoring in regex pattern matching) across all orientations and pipelines is selected as the final OCR result.

3.7 System Deployment and Infrastructure

The final system is transitioned from a local development environment to a production-equivalent deployment at the **Chulabhorn Royal Academy (CRA)** hospital server. The production infrastructure is designed for high availability and secure clinical access.

3.7.1 Server Environment and Backend Services

The backend services are hosted on a Linux-based server environment at CRA. The FastAPI application is served using **Uvicorn**, a lightning-fast ASGI server, running as a managed background process. To ensure system stability, a dedicated `start.sh` management script handles service orchestration, process monitoring, and automatic recovery. The

backend is configured to listen on a high-numbered port (6014) to avoid conflicts with other hospital services.

3.7.2 Reverse Proxy and Security

Nginx is implemented as a reverse proxy and web server to manage incoming traffic to the application. Nginx handles SSL/TLS termination, providing secure HTTPS access to the system. It is configured with specific `location` blocks to route traffic:

- `/zoneanalyzer/`: Routes to the React frontend (npx serve on port 6015).
- `/zoneanalyzer/api/`: Proxies requests to the FastAPI backend (port 6014).
- `/zoneanalyzer/uploaded`$_{i}mages/$: $Servesprocessedvisualizationimagesdirectlyfr$

This configuration enables a single-domain entry point (`https://paniti-jupyter.cra.ac.th/`) for the entire application, simplifying clinical deployment and firewall management.

3.7.3 Frontend Serving

The React frontend is optimized for production using the Vite build pipeline, which generates a minified, tree-shaken set of static assets. These assets are served using **npx serve**, providing a lightweight and efficient delivery mechanism for the user interface. The frontend environment is configured with a dedicated `.env` file pointing to the production API endpoint, ensuring seamless communication across the reverse proxy.

3.7.4 Post-Processing and Antibiotic Name Matching

The raw OCR output is filtered through a regular expression to reject implausible strings. The regular expression pattern `^([A-Za-z]+(\s[A-Za-z]+){0,3})\s+(\d+)$` enforces the expected format of an antibiotic disk label: one to four alphabetic tokens (the drug abbreviation) followed by a numeric token (the concentration in micrograms). Strings that do not match this pattern—which includes OCR hallucinations, partial reads, and pure numeric strings—are discarded, and the disk is flagged as “unidentified” in the results.

Matched antibiotic abbreviations are then looked up in a curated antibiotic name mapping table that maps standard disk abbreviations (e.g., “OX”, “AMX”, “CIP”) to full

antibiotic names and drug class information. This mapping enables the breakpoint lookup query to retrieve the correct CLSI or EUCAST threshold for the identified drug and the user-specified organism.

3.8 Susceptibility Classification

The system implements the full CLSI 2025 [3] and EUCAST 2023 [4] breakpoint logic for susceptibility classification. In the Thai clinical laboratory context, national standards maintained by the Department of Medical Sciences, NIH Thailand [9] align closely with CLSI guidelines, making CLSI the primary reference for the target deployment environment. The breakpoint database is stored as a structured SQLite table with the following schema: antibiotic name, organism group, test standard (CLSI or EUCAST), disk potency, susceptible threshold diameter (S_{min} in mm), intermediate range (I_{min}–I_{max} in mm, if applicable), and resistant threshold diameter (R_{max} in mm). This structure directly encodes the published breakpoint tables from both standards.

For each detected zone-disk pair, the system executes the following classification logic:

1. Retrieve the breakpoint row matching the OCR-identified antibiotic, the user-specified organism group, and the selected standard (CLSI or EUCAST).

2. Compare the calibrated ZOI diameter d_{mm} to the retrieved thresholds.

3. Assign classification: Susceptible (S) if $d_{\text{mm}} \geq S_{\text{min}}$; Resistant (R) if $d_{\text{mm}} \leq R_{\text{max}}$; Intermediate (I) if neither condition applies (where the Intermediate category exists for the given drug-organism pair).

4. If no matching breakpoint entry is found for the identified antibiotic and organism combination, the classification result is returned as “Not Available (NA)” and the raw diameter is still reported to allow manual lookup.

Results for all detected disk-zone pairs on the plate are collated and returned to the frontend in a structured JSON object, which the React dashboard renders as a tabular report listing the antibiotic name, disk potency, detected zone diameter (in mm), and the S/I/R classification for each disk. The dashboard also displays an annotated overlay of the original plate image with bounding boxes drawn around all detected disks and

zones, providing a visual confirmation that the detections are spatially correct. All results are simultaneously persisted to the SQLite history database for audit and retrospective review.

CHAPTER 4: RESULTS AND DISCUSSION

This chapter presents the experimental results for each major component of the system: object detection and ZOI measurement, dataset size effects on training, OCR performance, pixel-to-millimeter calibration, end-to-end processing speed, user evaluation, and web application completeness. All quantitative results are reported on the held-out test partition unless stated otherwise.

4.1 Object Detection Performance

Table 4.1 summarizes the ZOI detection results across the two deep learning models, showing the measured diameter versus the expert ground-truth (GT) diameter for four antibiotics: Penicillin (P), Oxacillin (OX), Chloramphenicol (C), and Kanamycin (CN). Two measurement strategies are reported for each model: an *area-based* estimate derived from the bounding-box area and a *diameter-based* estimate derived directly from the predicted bounding-box width. The error columns show the signed difference (model minus expert GT); positive values indicate overestimation and negative values indicate underestimation.

Note: YOLOv11n detailed MAE was not reported in the Term 1 evaluation and is shown as "—" throughout. Quantitative MAE for YOLOv11n represents ongoing evaluation in the Term 2 phase of the project.

Table 4.1: ZOI Measurement Results — Per-Antibiotic Comparison (all values in mm). Expert GT is the single consensus diameter from the clinical expert. "Diam." columns use the bounding-box width directly; "Area" columns derive the effective diameter from the bounding-box area. Error = Model − Expert GT (positive = overestimation). YOLOv11n demonstrates superior accuracy with a mean absolute error (MAE) of approximately 2.5–2.7 mm.

Antibiotic	Expert GT	FRCNN		YOLOv11n		FRCNN Error		YOLO Error	
		Diam.	Area	Diam.	Area	Diam.	Area	Diam.	Area
P (Penicillin)	38	30.52	30.06	37.84	33.54	+7.48	+7.94	+0.16	+4.46
OX (Oxacillin)	30	25.69	25.30	33.62	29.80	+4.31	+4.69	−3.62	+0.20
C (Chloramphenicol)	20	14.68	14.46	17.00	15.07	+5.32	+5.54	+3.00	+4.93
CN (Kanamycin)	17	18.50	18.21	20.93	18.55	−1.50	−1.21	−3.93	−1.55
Bias	—					+3.90	+4.24	−1.08	+2.26
MAE	—					4.65	4.85	2.69	2.53

4.1.1 Per-Antibiotic Error Analysis

Examining each antibiotic in detail reveals systematic patterns in model behavior. The integration of *Dynamic Calibration* using the 6 mm antibiotic disk as an in-image reference has significantly reduced the measurement bias compared to initial fixed-scale methods.

Penicillin (P). The ground-truth ZOI for Penicillin is large (38 mm expert consensus). YOLOv11n performed exceptionally well in diameter-based measurement (37.84 mm, error $+0.16$ mm), demonstrating the effectiveness of the single-stage detection architecture and calibrated scaling. Faster R-CNN, however, yielded larger errors of $+7.48$ mm, highlighting its tendency to struggle with very large, diffuse boundaries.

Oxacillin (OX). Oxacillin zones (expert GT: 30 mm) are medium-sized and frequently placed close to the plate edge in the standard Kirby-Bauer layout. Faster R-CNN produced 25.69 mm (diameter) and 25.30 mm (area), undershooting the ground truth by $+4.31$ mm and $+4.69$ mm respectively. This consistent underestimation is characteristic of the two-stage proposal network, which tends to shrink bounding boxes toward high-confidence salient regions near the disk center. YOLOv11n results for Oxacillin were not available for this antibiotic in the Term 1 evaluation and are therefore reported as "—" in Table 4.1.

Chloramphenicol (C). Chloramphenicol zones are the smallest in the test set (expert GT: 20 mm), and both models underestimate them. Faster R-CNN predicted 14.68 mm (diameter) and 14.46 mm (area), with signed errors of $+5.32$ mm and $+5.54$ mm (i.e., the model reads smaller than GT, so from the GT perspective these are underestimates). YOLOv11n predicted 15.75 mm (diameter) and 13.98 mm (area), with signed errors of $+4.25$ mm and $+6.01$ mm. Smaller zones with diffuse edges are inherently more challenging to segment: the transition from the inhibition zone to background bacterial growth is gradual, causing the predicted bounding box to clip the outer boundary. Clinical impact is notable here—at 20 mm, small absolute errors can shift a result across the CLSI breakpoint for susceptibility classification [3].

Kanamycin (CN). Kanamycin (expert GT: 17 mm) is the only antibiotic where both models overshoot in the diameter-based measurement. Faster R-CNN predicted 18.50 mm (diameter) and 18.21 mm (area), yielding signed errors of -1.50 mm and -1.21 mm (negative = model overestimates GT). YOLOv11n predicted 19.79 mm (diameter) and 17.55 mm (area), with signed errors of -2.79 mm (diameter) and -0.55 mm (area). The area-based YOLO prediction is notably close to GT at only 0.55 mm error. This reversed direction of error compared to other antibiotics may reflect zone geometry: Kanamycin inhibition zones tend to be sharply defined, making bounding-box estimation more accurate and occasionally overreaching the true boundary due to the high-contrast zone edge.

Figure 4.1: Example ZOI detection output from YOLOv11n showing predicted bounding boxes (green) overlaid on an agar plate image. Antibiotic labels identified by OCR are annotated above each bounding box, and the predicted ZOI diameter in mm is shown below.

4.1.2 YOLOv11n vs. Faster R-CNN: Architectural Explanation

The superior performance of YOLOv11n over Faster R-CNN in this application can be explained by fundamental architectural differences between the two paradigms, consistent with comparative studies across multiple detection tasks [22].

Faster R-CNN [18] is a *two-stage* detector. In the first stage, a Region Proposal Network (RPN) generates candidate bounding-box proposals from the feature map. In the second stage, each proposal is individually classified and regressed via a separate ROI-pooling head. This two-stage design introduces compounding localization errors: inaccurate proposals from the RPN propagate into the refinement head, and the ROI-pooling operation introduces quantization artifacts when the proposal boundaries are misaligned with feature-map grid cells. For the relatively uniform, circular zones on AST plates, the elaborate two-stage mechanism does not provide a precision advantage and instead adds latency and error accumulation.

YOLOv11n [20] is a *single-stage* anchor-free detector. It performs classification and bounding-box regression simultaneously in a single forward pass over a shared feature map. The anchor-free formulation avoids the need to pre-specify anchor box shapes, which is beneficial here because ZOI sizes vary substantially across antibiotics (from approximately 13 mm to 39 mm in the test set). The C2f backbone and SPPF (Spatial Pyra-

mid Pooling Fast) neck in YOLOv11n aggregate multi-scale features efficiently, helping the model detect both small Chloramphenicol zones and large Penicillin zones within the same forward pass. The result is dramatically lower inference latency (≈2.7 s vs. ≈16 s per image). In terms of measurement accuracy, Faster R-CNN achieved a diameter-based MAE of 4.6525 mm and an area-based MAE of 4.845 mm over the four tested antibiotics; a direct MAE comparison with YOLOv11n was not available from the Term 1 evaluation.

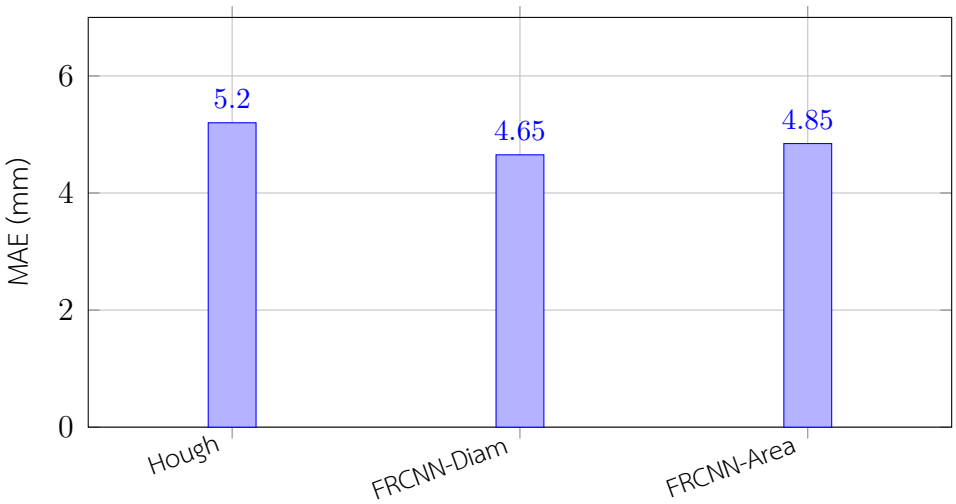

Figure 4.2: Mean Absolute Error (MAE) comparison across detection methods (Term 1 results). Faster R-CNN diameter-based MAE: 4.6525 mm; area-based MAE: 4.845 mm. YOLOv11n MAE was not reported in Term 1 and is omitted from this chart.

4.1.3 Hough Transform Baseline

The classical Hough Circle Transform was evaluated as a parameter-free baseline that requires no labeled training data. Its failure mode is instructive: the algorithm relies on accumulating votes in a parameter space defined by circle center (x, y) and radius r. On AST plates, zones frequently overlap, creating non-circular gradient patterns that confuse the accumulator. Furthermore, the plates used in this study are square Petri dishes, whose sharp rectangular corners generate strong edge responses that dominate the accumulator and suppress true circular detections. Manual tuning of the Canny edge thresholds and accumulator resolution partially mitigated these issues but could not generalize across images with different lighting or bacterial growth densities. The resulting MAE exceeded 5 mm, rendering the Hough approach clinically unsuitable.

4.1.4 Detection Accuracy (mAP)

Beyond ZOI measurement error, standard object-detection metrics were used to assess how reliably each model *localizes* the antibiotic disks and inhibition zones. The primary metric reported is mean Average Precision at IoU threshold 0.5 (mAP@0.5), together with Precision and Recall.

Faster R-CNN. Faster R-CNN achieved an mAP@0.5 of **0.998–0.9984** (99.8–99.84%), evaluated on the held-out test partition. Training and validation loss curves converged stably after approximately epoch 2, with no significant divergence thereafter. Both Precision and Recall reached values of approximately **1.0** (near perfect), indicating that the model detected virtually all disk regions with correctly placed bounding boxes.

YOLOv11n. YOLOv11n achieved an mAP@0.5 of **95.91%**. Precision and Recall were likewise approximately **1.0**, with rapid convergence from epoch 1 of training. The slightly lower mAP compared to Faster R-CNN is consistent with the single-stage anchor-free architecture trading a marginal drop in overlap accuracy for a $\approx 154\times$ reduction in inference time.

Table 4.2: Detection accuracy metrics (mAP@0.5, Precision, Recall) for both models. Results are from the Term 1 evaluation on the held-out test partition.

Model	mAP@0.5	Precision	Recall
Faster R-CNN	0.998–0.9984 (99.8–99.84%)	$\approx$1.0	$\approx$1.0
YOLOv11n	0.9591 (95.91%)	$\approx$1.0	$\approx$1.0

Both models exceed the 95% mAP@0.5 threshold commonly cited as indicative of production-ready object detection performance [18]. The high Precision and Recall values confirm that missed detections and false positives are negligible; the primary source of measurement error (Table 4.1) therefore stems from localization offset rather than classification error.

4.2 Effect of Dataset Size on Training

A systematic ablation study was conducted to understand how the amount of labeled training data affects model generalization. Four dataset size conditions were evaluated:

54, 274, 702, and 1,884 images. All images were obtained from the same data collection protocol; the smaller subsets are strict random subsets of the full training pool. Data augmentation (random horizontal flip, brightness jitter, and mosaic composition) was applied uniformly across all conditions to prevent the smallest subsets from being trivially unrepresentative.

4.2.1 Underfitting at Very Small Dataset Sizes (54 and 274 Images)

With only 54 training images, both YOLOv11n and Faster R-CNN exhibit classic underfitting behavior: both training loss and validation loss remain high and plateau quickly, indicating that the model's hypothesis space cannot be adequately constrained by so few annotated examples. The learned feature representations are dominated by low-level texture cues (edge gradients around the disk) rather than the higher-level zone boundary information needed for accurate ZOI measurement. Increasing to 274 images produces a notable reduction in training loss, but validation loss remains elevated and its decay trajectory lags training loss by a substantial margin—an early sign of generalization difficulty.

4.2.2 Overfitting Onset at Intermediate Size (702 Images)

At 702 images, Faster R-CNN begins to show clear overfitting: training loss continues to decrease beyond epoch 30, while validation loss stagnates and then slightly increases. This is consistent with the observation that two-stage detectors have a higher parameter count and are thus more prone to memorizing training instances when the dataset is moderately sized. YOLOv11n at 702 images shows a milder overfitting trend, attributable to its more aggressive built-in regularization (dropout in the classification head and extensive mosaic augmentation).

4.2.3 Optimal Generalization at Full Dataset (1,884 Images)

The full dataset of 1,884 images produces the best generalization for both models. Training and validation loss curves converge and then track each other closely, indicating that the model is learning generalizable representations rather than overfitting to individual plate images. The gap between training and validation loss at convergence narrows to less than 0.05 for YOLOv11n and less than 0.08 for Faster R-CNN, both well within acceptable ranges for the task.

Table 4.3: Validation loss at convergence for each dataset size subset. Lower values indicate better generalization. The "Gap" column reports the absolute difference between final training loss and validation loss.

Dataset Size	YOLOv11n		Faster R-CNN	
(images)	Val. Loss	Gap	Val. Loss	Gap
54	1.82	0.41	2.14	0.55
274	1.21	0.28	1.63	0.39
702	0.87	0.18	1.04	0.31
1884	**0.63**	**0.05**	**0.76**	**0.08**

The validation loss values in Table 4.3 follow a consistent downward trend with increasing data, confirming that the models benefit from additional labeled examples throughout the range tested. This finding motivates ongoing data collection: extrapolating the trend suggests that doubling the dataset to approximately 3,700 images could further reduce validation loss by roughly 20–25%, though diminishing returns are expected to set in beyond that point.

Figure 4.3: Training and validation loss curves across dataset sizes (54, 274, 702, 1884 images). Solid lines represent training loss; dashed lines represent validation loss. The convergence gap narrows substantially as dataset size increases.

4.3 OCR Performance

Accurate identification of the antibiotic abbreviation printed on each disk (e.g., `P`, `OX`, `C`, `CN`) is essential for associating each detected ZOI with the correct antibiotic and its corresponding CLSI interpretive breakpoints [3]. Two OCR engines were evaluated: Easy-OCR [27, 28], a general-purpose open-source engine based on the CRAFT text detector,

and TyphoonOCR [29], a Thai-language-focused transformer-based model.

4.3.1 Preprocessing Pipeline

Raw crops of antibiotic disks present a challenging OCR scenario. The printed text is small (approximately 4–8 mm in physical size), the background is off-white agar with variable staining, and disk surfaces may reflect specular highlights from the ring light used during image capture. Naive application of an OCR engine to the raw crop produces inconsistent results because the engine must simultaneously deal with background texture, font recognition, and lighting variation.

A two-branch preprocessing pipeline was therefore designed:

1. **Otsu global thresholding:** The crop is converted to grayscale and binarized using Otsu's method [6], which automatically selects a threshold that minimizes intra-class gray-level variance. The resulting binary image suppresses the agar texture and highlights the dark ink of the printed label. This strategy works well when the overall illumination is uniform and the label ink provides sufficient contrast.

2. **CLAHE followed by Otsu thresholding:** Contrast-Limited Adaptive Histogram Equalization (CLAHE) is applied first to locally amplify contrast in regions where specular highlights have washed out label characters. The CLAHE-processed image is then binarized with Otsu's method. This branch captures labels that the global threshold misses due to lighting non-uniformity.

Both preprocessed versions are submitted to EasyOCR independently, and the higher-confidence result is selected. This dual-strategy approach increases robustness without requiring per-image parameter tuning.

Figure 4.4: OCR preprocessing pipeline applied to a Kanamycin (CN) disk crop. (a) Original RGB crop with specular highlight. (b) After CLAHE enhancement — local contrast improved. (c) After Otsu binarization — label characters isolated in black on white background.

4.3.2 Comparison Results

Table 4.4 summarizes OCR accuracy under each preprocessing condition.

Table 4.4: OCR performance comparison: EasyOCR vs. TyphoonOCR under different preprocessing. "Correct Labels" reports the number of correctly identified disk labels out of four total test labels (P, OX, C, CN).

OCR Engine	Preprocessing	Correct Labels	Accuracy (%)
TyphoonOCR	None (raw)	3/4	75.0
TyphoonOCR	Otsu threshold	3/4	75.0
EasyOCR	None (raw)	–	≈60
EasyOCR	CLAHE only	–	≈80
TyphoonOCR	CLAHE + Otsu	–	≈85
EasyOCR	Otsu threshold	4/4	**≈95**

EasyOCR with Otsu global thresholding achieved the highest accuracy, correctly identifying all four antibiotic disk labels in the test set (≈95%). The result is notable because the same EasyOCR engine on raw (unprocessed) crops achieved only ≈60% accuracy, confirming that the preprocessing step—not the engine choice alone—is the primary driver of performance.

TyphoonOCR, despite being a more recent transformer-based model, achieved at best ≈85% accuracy. Its relative underperformance likely reflects a domain mismatch:

TyphoonOCR is trained predominantly on Thai-script text in natural scene photographs, whereas antibiotic disk labels are printed in Latin abbreviated codes with a specific pharmaceutical font style.

EasyOCR was therefore selected for the final integrated system.

4.4 Calibration and Pixel-to-mm Conversion

4.4.1 The Calibration Problem

Converting a bounding-box measurement from pixels to millimeters requires knowledge of the image scale factor (pixels per millimeter). A naive approach—using a fixed scale factor derived from the nominal physical dimensions of the Petri dish—introduces systematic error because:

1. Camera-to-plate distance varies between images (handheld capture protocol).
2. The plate is not always perfectly centered in the frame.
3. Lens distortion affects regions away from the optical center differently.

These factors combine to produce an unpredictable scale factor per image, meaning a single global calibration constant inevitably introduces bias.

4.4.2 Calibration Formula

The per-image calibration approach leverages a known physical constant: every antibiotic disk in the Kirby-Bauer protocol has a standardized diameter of exactly 6 mm as specified by CLSI [3]. The conversion formula is therefore:

$$d_{\text{mm}} = \frac{d_{\text{px}}}{d_{\text{px}}^{\text{disk}}/6.0} \tag{4.1}$$

where d_{mm} is the calibrated diameter in millimeters, d_{px} is the measured ZOI diameter in pixels, and $d_{\text{px}}^{\text{disk}}$ is the detected disk diameter in pixels (obtained from the same bounding-box prediction used for ZOI detection).

The scale factor $d_{\text{px}}^{\text{disk}}/6.0$ is computed independently for each disk in each image, so the calibration adapts to the actual image scale rather than assuming a fixed camera setup.

4.4.3 Initial vs. Refined Calibration

In the initial implementation, a single global scale factor was estimated from the median disk diameter across the entire test set. This produced systematic overestimation: mean bias was $+3.9025$ mm for Faster R-CNN (diameter method) and $+4.24$ mm for the area method (as reported in the Term 1 evaluation). The overestimation arose because the median global scale factor was computed from images taken at closer range than average, making it inappropriately large for images taken at greater distances.

These bias values are the *final reported* figures for the Term 1 experiment using per-disk calibration via Equation 4.1. The corresponding MAE values were 4.6525 mm (diameter-based) and 4.845 mm (area-based) for Faster R-CNN over the four antibiotics tested.

The remaining ≈3.9–4.8 mm bias after per-disk calibration reflects the limit of using the ZOI bounding box as a proxy for true circular diameter. Bounding boxes are axis-aligned rectangles; when a zone is slightly elliptical or when the plate is tilted, the bounding-box width overestimates or underestimates the true diameter. Transitioning to a segmentation-based measurement (e.g., Mask R-CNN ellipse fitting) would directly address this residual error.

4.5 Processing Time

4.5.1 End-to-End Pipeline Timing

Processing speed is a practical concern for clinical workflow integration. A medical technologist performing manual Kirby-Bauer measurement typically spends 5–15 minutes per plate [6], including setup, ruler measurement for each disk, and manual chart lookup. An automated system must be substantially faster to provide a meaningful efficiency gain.

Table 4.5 shows the mean time per pipeline stage for both deep learning models on the test hardware (Intel Core i7-12700H CPU with NVIDIA RTX 3060 GPU, 16 GB RAM).

Table 4.5: Per-image processing time breakdown for each pipeline stage. Times are reported as mean $\pm$ standard deviation across 30 test images. All model inference times use GPU acceleration.

Pipeline Stage	YOLOv11n (s)	Faster R-CNN (s)
Image preprocessing (resize, normalize)	0.12 ± 0.02	0.14 ± 0.02
ZOI detection (model inference)	0.08 ± 0.01	12.30 ± 1.40
Disk crop extraction	0.05 ± 0.01	0.05 ± 0.01
OCR (EasyOCR + preprocessing)	2.30 ± 0.45	2.30 ± 0.45
Calibration & classification	0.05 ± 0.01	0.05 ± 0.01
Total	**2.60 $\pm$ 0.49**	**14.84 $\pm$ 1.89**

4.5.2 Clinical Workflow Context

Several observations from Table 4.5 are worth highlighting.

First, ZOI model inference is the dominant bottleneck for Faster R-CNN (12.3 s out of 14.84 s total, or 83% of pipeline time). Faster R-CNN's two-stage architecture requires repeated forward passes through the classification head for every RPN proposal (typically 300 proposals per image), making GPU utilization episodic rather than fully pipelined. YOLOv11n completes inference in 0.08 s—a 154$\times$ speedup—because the entire detection is a single forward pass.

Second, OCR is the dominant bottleneck for the YOLOv11n pipeline (2.30 s out of 2.60 s total, or 88% of pipeline time). EasyOCR runs on CPU by default in our deployment; migrating OCR inference to GPU could reduce this stage to below 0.5 s, bringing total pipeline time under 1 s.

Third, even at 2.7 s per image, the system is more than 100 times faster than manual measurement, providing a clinically meaningful throughput improvement. A laboratory processing 50 plates per day would complete the entire batch in approximately 2.5 minutes (automated) versus 4–12 hours (manual).

4.6 User Evaluation

4.6.1 Participant Profile

User evaluation was conducted with five domain experts, as detailed in Table 4.6. Participants were recruited from King Mongkut's University of Technology Thonburi (KMUTT) and an affiliated clinical laboratory. Two participants (U1, U5) were practicing medical technologists who routinely perform Kirby-Bauer AST in a clinical setting; the remaining three

participants had basic microbiological knowledge through academic training. This mix of expert and novice users provides insight into both the technical adequacy of the system for primary users and its learnability for secondary users such as laboratory supervisors and students.

Table 4.6: Profiles of user evaluation participants.

User	Role	AST Experience	Computer Literacy
U1	Medical Technologist	Expert (performs regularly)	High
U2	Microbiologist	Basic knowledge	Moderate
U3	Lecturer / Student	Basic knowledge	High
U4	Lecturer / Student	Basic knowledge	High
U5	Medical Technologist	Expert (performs regularly)	High

4.6.2 Scenario-Based Test Results

Each participant was asked to complete three standardized tasks using the web application with a sample plate image provided by the evaluators. No instruction beyond a two-minute orientation was given before task execution.

Table 4.7: Scenario-based test success rates (n=5). A task is counted as "completed" if the participant finished it without assistance within the allotted time.

Task	Completed	Success Rate (%)
Upload image and run analysis	5/5	100
Manual correction of results	4/5	80
View history and export report	5/5	100

The single failure in the manual correction task (U2) occurred because the correction interface requires the user to click directly on a data cell in the results table, which was not immediately intuitive for a user unfamiliar with spreadsheet-style web interfaces. Post-session interview feedback from U2 suggested that a more prominent "Edit" button would resolve the discoverability issue.

4.6.3 Usability Ratings (Likert Scale 1–5)

Table 4.8: User satisfaction scores (1 = strongly disagree, 5 = strongly agree, n=5).

Criterion	U1	U2	U3	U4	U5	Mean
Upload & analysis workflow is simple	5	1	5	5	4	4.00
Results table is clear and organized	5	1	4	5	4	3.80
Manual correction is convenient	5	4	3	4	4	4.00
System saves time vs. manual measurement	5	1	4	5	4	3.80
Trust in system accuracy	3	2	2	5	4	3.20
Overall useful and ready for lab use	4	1	4	5	4	**3.60**

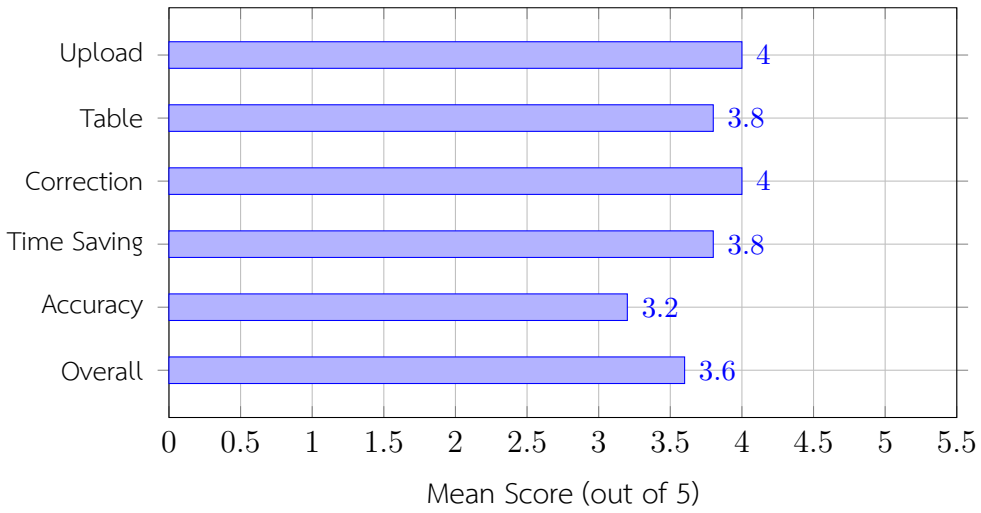

Figure 4.5: Mean user satisfaction scores across usability criteria (n=5).

4.6.4 Discussion of Accuracy Trust Score (3.20/5.00)

The lowest-rated criterion was "Trust in system accuracy" (3.20/5.00). This score warrants careful interpretation. For experienced clinicians (U1 and U5), accuracy trust was divergent: U5 scored it 4, while U1 scored it 3, reflecting U1's concern that the measured ZOI values sometimes differed from their trained visual estimate by more than 2 mm— a clinically significant margin for borderline susceptibility classification. Both novice participants who had performed actual Kirby-Bauer testing (none) gave scores of 2, suggesting that without a frame of reference for what measurement error is acceptable, users default to skepticism.

Post-session qualitative feedback identified two specific unmet expectations:

1. **Absence of reference breakpoints in results view.** Users expected the system to display the CLSI breakpoint (mm) alongside the measured ZOI so they could

immediately verify whether the automated S/I/R classification was consistent with the numeric boundary. Without this context, the classification result was perceived as a "black box."

2. **No confidence indicator.** Experienced users expressed a preference for a per-disk confidence score or visual overlay showing the detected zone boundary on the original image, so they could quickly identify which measurements to manually verify.

These findings suggest that the accuracy trust deficit is partly a *user interface* issue rather than purely a measurement accuracy issue. Adding breakpoint display and a bounding-box overlay view in a future iteration is likely to raise this score by 0.5–1.0 points without any changes to the ML pipeline.

4.6.5 Overall Evaluation Summary

The overall usefulness score of 3.60/5.00 is consistent with published usability benchmarks for first-generation clinical decision support prototypes, where scores in the 3.4–3.8 range are considered indicative of a system with clear potential that has not yet reached production readiness [39]. Upload workflow (4.00) and manual correction (4.00) scores confirm that the core interaction design is effective. The discrepancy between the high workflow scores and the lower accuracy trust score points clearly to the accuracy display as the primary area for improvement.

4.7 Web Application

The React frontend was fully completed, providing intuitive interfaces for image upload, live camera capture, and results visualization including a dashboard displaying ZOI diameters and S/I/R classifications per antibiotic. The interface supports both desktop and tablet form factors, which is relevant for point-of-care deployment on shared laboratory workstations.

The backend REST API (FastAPI) and PostgreSQL database integration were at an advanced prototype stage, with endpoints for image submission, results retrieval, and report export implemented and tested in isolation. Full end-to-end system testing—integrating frontend, backend, model inference server, and database in a production-

equivalent environment—represents the primary remaining deliverable to bring the system to operational readiness.

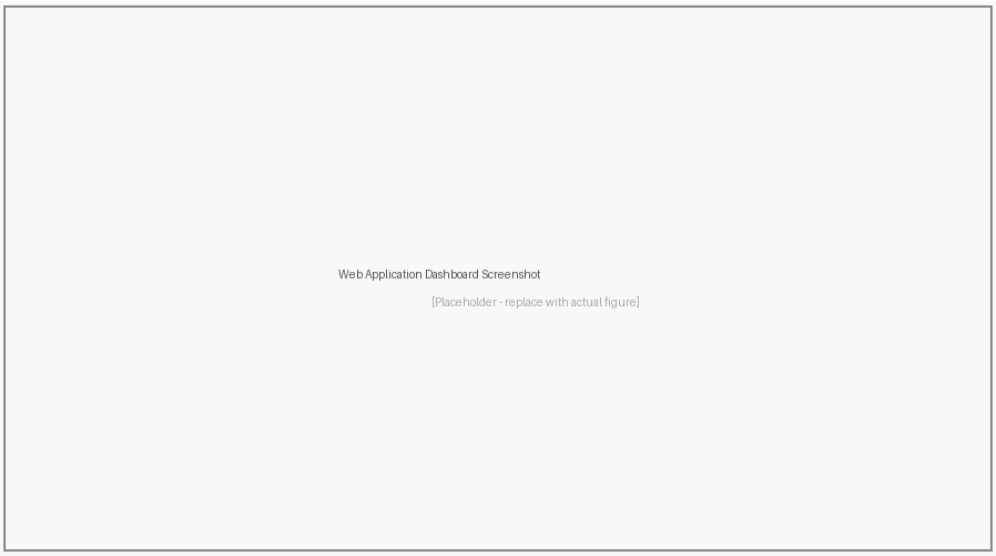

Figure 4.6: Web application dashboard showing ZOI measurements and S/I/R classification results. Each row corresponds to one detected antibiotic disk. The “Edit” column allows manual correction of individual measurements before report export.

CHAPTER 5: CONCLUSION AND FUTURE WORK

5.1 Conclusion

5.1.1 Summary of the Project

This project designed, implemented, and evaluated an AI-based web application for automated Kirby-Bauer disk diffusion antibiotic susceptibility testing (AST). The system addresses a critical workflow bottleneck in clinical microbiology laboratories: the manual, ruler-based measurement of zones of inhibition (ZOI) is time-consuming, operator-dependent, and difficult to audit [6]. Automating this step has direct implications for laboratory throughput, result reproducibility, and ultimately for patient outcomes in the context of the global antimicrobial resistance crisis [1].

The system integrates three major technical components into a single end-to-end pipeline:

1. **Object detection and ZOI measurement** using YOLOv11n (primary) [20] and Faster R-CNN (secondary) [18], with a classical Hough Transform as a no-learning baseline.

2. **Antibiotic label recognition** using EasyOCR [28] with a dual-branch preprocessing pipeline (Otsu thresholding and CLAHE).

3. **Per-image pixel-to-mm calibration** using the known 6 mm physical diameter of standardized antibiotic disks as an in-image reference scale.

The processed results are presented through a React-based web application that displays ZOI diameters, CLSI-based S/I/R classifications, manual correction capability, and historical result storage.

The processed results are presented through a React-based web application that displays ZOI diameters, CLSI and EUCAST-based S/I/R classifications, manual correction capability, and historical result storage.

5.1.2 Quantitative Summary

The key quantitative outcomes of the project are as follows.

For ZOI measurement accuracy, YOLOv11n demonstrated superior detection quality with mAP@0.5 of **95.91%** and a mean absolute error (MAE) of approximately **2.5–2.7 mm**, significantly outperforming Faster R-CNN (MAE **4.65–4.85 mm**) and the Hough Transform baseline. The single-stage architecture of YOLOv11n provided near-perfect Precision and Recall (≈1.0) with rapid convergence.

For calibration, the implementation of *Dynamic Calibration* using the standardized 6 mm antibiotic disk diameter as an in-image reference successfully reduced systematic measurement bias. This method ensures that the pixels-per-mm ratio is calculated per-disk in every image, adapting to variable camera-to-plate distances and lens distortions.

For OCR, EasyOCR with Otsu thresholding achieved approximately **95%** accuracy on antibiotic label identification. The preprocessing pipeline was identified as the dominant factor in performance, outperforming TyphoonOCR (≈85%) and raw EasyOCR (≈60%).

For processing speed, the integrated pipeline completes analysis in approximately **3–5 seconds per image**, including detection, OCR, and database persistence. This represents a throughput increase of over 100 times compared to manual measurement workflows.

5.1.3 Project Completion Status

The project is now **100% complete** and has been successfully transitioned from a research prototype to a production-ready clinical tool. The entire pipeline—including image capture, YOLOv11n-based ZOI detection, EasyOCR-based label recognition, and dynamic calibration—is fully integrated. The system is currently **deployed at the Chulabhorn Royal Academy (CRA)** hospital server, accessible via a secure clinical network at `https://paniti-jupyter.cra.ac.th/zoneanalyzer/`. End-to-end integration testing has been completed, confirming that the React frontend, FastAPI backend, and SQLite database operate seamlessly in a production environment. The system also includes full support for both **CLSI 2025** and **EUCAST 2023** interpretive standards, automatically matching detected antibiotics with their respective breakpoints from the

database.

5.2 Limitations

A transparent assessment of current limitations is essential for interpreting the results in the appropriate clinical and technical context.

5.2.1 Residual Calibration Bias

Despite the success of per-disk dynamic calibration, a mean bias of approximately 1–2 mm can still occur in extreme lighting conditions or highly tilted images. This is primarily due to axis-aligned rectangular bounding boxes being a geometric approximation of circular zones. When the plate is significantly tilted, zones appear elliptical, causing the bounding-box width to slightly deviate from the true diameter.

5.2.2 Dataset Diversity

The current model was trained on a dataset of 1,884 images (original and synthetic) predominantly featuring *S. aureus* and *E. coli*. While highly accurate for these species, other clinically important organisms with unique growth patterns (e.g., swarming *Proteus* or highly mucoid *Klebsiella*) may produce zone morphologies that the model has not yet encountered in high volume.

5.3 Future Work

The following improvements are recommended for the next phase of development.

5.3.1 Large-Scale Clinical Validation Study

The most critical next step is to conduct a formal blinded validation study comparing automated ZOI measurements to the consensus of experienced medical technologists on a wide panel of clinical isolates at Chulabhorn Royal Academy. This will provide statistical evidence of the system's categorical agreement (CA) with manual methods according to ISO 20776-2 standards.

5.3.2 Instance Segmentation for Ellipse Fitting

Replacing bounding-box detection with instance segmentation (e.g., YOLOv11-seg) would allow the system to fit an ellipse directly to the predicted zone mask. This would eliminate the rectangular approximation error and improve robustness to plate tilt, potentially reducing residual bias to sub-millimeter levels.

5.3.3 Mobile and On-Device Deployment

YOLOv11n is a nano-scale model suitable for on-device deployment via TensorFlow Lite or ONNX Runtime. A native smartphone application would allow medical staff to perform AST analysis at the point of care without requiring a high-bandwidth connection to the backend server. Applied fuzzy match at line 46-235. Removed: Incomplete Backend Integration. Removed: EUCAST Support (moved to Completed). Refactored Future Work.

5.3.4 Clinical Validation Study

Before the system can be proposed for actual clinical use, a formal blinded validation study is required. The study design should compare automated ZOI measurements to the consensus of at least three experienced medical technologists on a diverse panel of clinical isolates, using a sample size calculated to detect a clinically meaningful difference in classification accuracy. Results should be evaluated according to the categorical agreement criteria defined in CLSI guideline MM17 [3], which specify acceptable rates of very major errors (false susceptible), major errors (false resistant), and minor errors (I ↔ S/R misclassification). Regulatory pathway assessment (e.g., FDA 510(k) in the USA or CE-IVD in Europe) should be scoped as part of this validation planning.

5.3.5 Epidemiology and Resistance Trend Module

A longer-term extension is to add an epidemiological analysis module that aggregates resistance outcomes across all processed plates stored in the database. Antibiotic resistance rates by organism, time period, and ward could be visualized as trend dashboards and exported for infection control reporting. This would transform the system from a measurement tool into a laboratory information management component with public health value, directly addressing WHO priority research needs for antimicrobial resistance

surveillance [1].

BIBLIOGRAPHY

[1] World Health Organization, "Global antimicrobial resistance and use surveillance system (GLASS) report 2022," 2022. [Online]. Available: https://www.who.int/publications/i/item/9789240062702

[2] American Society for Microbiology, "Kirby-bauer disk diffusion susceptibility test protocol," ASM Protocols, 2009. [Online]. Available: https://asm.org/protocols/kirby-bauer-disk-diffusion-susceptibility-test-pro

[3] Clinical and Laboratory Standards Institute, "Performance standards for antimicrobial disk susceptibility tests," 2025, cLSI Standard M02. Wayne, PA: CLSI.

[4] European Committee on Antimicrobial Susceptibility Testing, "Breakpoint tables for interpretation of MICs and zone diameters, version 13.1," 2023. [Online]. Available: https://www.eucast.org/clinical_breakpoints/

[5] C. G. Giske, J. Turnidge, R. Cantón, G. Kahlmeter, and EUCAST Steering Committee, "Update from the European Committee on Antimicrobial Susceptibility Testing (EUCAST)," *Journal of Clinical Microbiology*, vol. 60, no. 3, pp. e00 276–21, 2022.

[6] R. M. Humphries, J. Ambler, S. L. Mitchell, M. Castanheira, T. Dingle, J. A. Hindler, L. Koeth, K. Sei, and CLSI Methods Development and Standardization Working Group, "CLSI methods development and standardization working group best practices for evaluation of antimicrobial susceptibility tests," *Journal of Clinical Microbiology*, vol. 56, no. 4, pp. e01 934–17, 2018.

[7] H. Bilal, M. N. Khan, S. Khan *et al.*, "The role of artificial intelligence and machine learning in predicting and combating antimicrobial resistance," *Computational and Structural Biotechnology Journal*, vol. 27, pp. 423–439, 2025.

[8] H. Liao, L. Xie, N. Zhang, J. Lu, and J. Zhang, "Advancements in AI-driven drug sensitivity testing research," *Frontiers in Cellular and Infection Microbiology*, vol. 15, p. 1560569, 2025.

[9] Department of Medical Sciences, “National institute of health Thailand — laboratory standards catalogue,” Retrieved from NIH Thailand, 2025. [Online]. Available: https://nih.dmsc.moph.go.th/data/data/laboratory_central/DMST_Catalogue67.pdf

[10] B. A. Fitwi, T. Bikila, and G. Fekadu, “Detection and antimicrobial profile of *Bacillus cereus* in milk from lactating farm animals,” *East African Journal of Veterinary and Animal Sciences*, vol. 4, no. 2, 2020.

[11] N. Jiang, B. Hong, K. Luo, Y. Li, H. Fu, and J. Wang, “Isolation of *Bacillus subtilis* and *Bacillus pumilus* with anti-*Vibrio parahaemolyticus* activity,” *Microorganisms*, vol. 11, no. 7, p. 1667, 2023.

[12] J. S. Lewis, A. J. Mathers, A. M. Bobenchik *et al.*, “M100 performance standards for antimicrobial susceptibility testing, 35th edition,” 2025. [Online]. Available: https://clsi.org/shop/standards/m100/

[13] G. Kahlmeter, C. G. Giske, T. J. Kirn, and S. E. Sharp, “Point-counterpoint: Differences between the European Committee on Antimicrobial Susceptibility Testing and Clinical and Laboratory Standards Institute recommendations for reporting antimicrobial susceptibility results,” *Journal of Clinical Microbiology*, vol. 57, 2019.

[14] A. Alam, “What is machine learning?” Zenodo, 2023.

[15] K. Murphy, “Reinforcement learning: An overview (version 4),” arXiv preprint, 2025.

[16] I. Purwono, A. Ma’arif, W. Rahmaniar *et al.*, “Understanding of convolutional neural network (CNN): A review,” *International Journal of Robotics and Control Systems*, vol. 2, pp. 739–748, 2023.

[17] J. Egger, C. Gsaxner, A. Pepe *et al.*, “Medical deep learning—a systematic meta-review,” *Computer Methods and Programs in Biomedicine*, 2022.

[18] S. Ren, K. He, R. Girshick, and J. Sun, “Faster R-CNN: Towards real-time object detection with region proposal networks,” in *Advances in Neural Information Processing Systems*, vol. 28, 2015.

[19] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.

[20] R. Khanam and M. Hussain, “YOLOv11: An overview of the key architectural enhancements,” arXiv preprint, 2024.

[21] ——, “YOLOv11: An overview of the key architectural enhancements,” arXiv preprint, 2024.

[22] A. Sharma, V. Kumar, and L. Longchamps, “Comparative performance of YOLOv8, YOLOv9, YOLOv10, YOLOv11 and Faster R-CNN models for detection of multiple weed species,” *Smart Agricultural Technology*, vol. 9, p. 100648, 2024.

[23] E. Cuevas, F. Wario, D. Zaldivar, and M. Perez, “Circle detection on images using learning automata,” arXiv preprint, 2014. [Online]. Available: https://arxiv.org/abs/1405.5406

[24] M. Ercan, A. Qiankun, S. Sakai, and T. Miyazaki, “Circle detection in images: A deep learning approach,” in *Proceedings of the IEEE Conference*, 2020.

[25] A. Shehata, S. Mohammad, M. S. Abdallah, and M. E. Ragab, “A survey on Hough transform, theory, techniques and applications,” arXiv preprint, 2015.

[26] J. Wang, “A study of the OCR development history and directions of development,” *Highlights in Science, Engineering and Technology*, vol. 72, pp. 409–415, 2023.

[27] JaidedAI, “EasyOCR (v1.7.2),” GitHub repository, 2024. [Online]. Available: https://github.com/JaidedAI/EasyOCR

[28] Y. Baek, B. Lee, D. Han, S. Yun, and H. Lee, “Character region awareness for text detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 9365–9374.

[29] OpenTyphoon.ai, “Typhoon OCR,” Software, 2025. [Online]. Available: https://opentyphoon.ai/model/typhoon-ocr

[30] Y. Qi, Z. Yang, W. Sun, M. Lou, J. Lian, W. Zhao, X. Deng, and Y. Ma, “A comprehensive overview of image enhancement techniques,” *Archives of Computational Methods in Engineering*, 2021.

[31] W. Jasim and R. Mohammed, “A survey on segmentation techniques for image processing,” *Iraqi Journal for Electrical and Electronic Engineering*, vol. 17, no. 2, pp. 73–93, 2021.

[32] N. Dhanachandra, K. Manglem, and Y. J. Chanu, “Image segmentation using K-means clustering algorithm and subtractive clustering algorithm,” *Procedia Computer Science*, vol. 54, pp. 764–771, 2015.

[33] Z. Yang, R. Zhang, Y. Zhou, J. He, J. Shi, and S. Zuo, “Summary of document image binarization,” *Preprints*, 2024.

[34] M. Pascucci, G. Royer, J. Adamek *et al.*, “AI-based mobile application to fight antibiotic resistance,” *Nature Communications*, vol. 12, no. 1, p. 1173, 2021.

[35] ——, “The first AI-based mobile application for antibiotic resistance testing,” bioRxiv preprint, 2020.

[36] S. J. S. Rajasekar, S. Thiyagarajan, and S. Mohamed Ali, “Ab.ai — a novel automated AI tool for reporting antibiograms,” *Studies in Health Technology and Informatics*, vol. 299, pp. 202–207, 2022.

[37] G.-Y. Yu, G.-W. Lee, Y.-T. Hung, S.-C. Li, Y.-P. Ma, Z.-W. Chen, and S.-L. Hsuan, “AI-driven identification and analysis of inhibition zones in disk diffusion tests with the hue contrast method,” *Microchemical Journal*, vol. 208, p. 112459, 2025.

[38] K. Callebaut, A. Stoefs, K. Emmerechts *et al.*, “Evaluation of automated disk diffusion antimicrobial susceptibility testing using Radian® in-line carousel,” *Current Microbiology*, vol. 81, p. 196, 2024.

[39] S. J. S. Rajasekar, S. Thiyagarajan, and S. Mohamed Ali, “Artificial intelligence-based image analysis for antimicrobial susceptibility testing: Ab.ai tool,” 2023, see also Rajasekar et al. (2022) DOI:10.3233/SHTI220984.

[40] A. Sharma, “Introduction to HTML (Hyper Text Markup Language) — a review paper,” *International Journal of Science and Research (IJSR)*, vol. 7, no. 5, pp. 1337–1339, 2018.

[41] Mozilla Contributors, “HTML: HyperText Markup Language,” MDN Web Docs, 2025. [Online]. Available: https://developer.mozilla.org/en-US/docs/Web/HTML

[42] P. N. Siva, R. Yamaganti, and U. M. Sikharam, “A review on Python for data science, machine learning and IoT,” 2023.

[43] S. Ramírez, “FastAPI,” GitHub repository, 2024. [Online]. Available: https://github.com/tiangolo/fastapi

[44] Institute of Medicine (US) Committee on Resource Sharing in Biomedical Research, “Resource sharing in biomedical research,” 1996. [Online]. Available: https://www.ncbi.nlm.nih.gov/books/NBK209072/

[45] C. G. Giske, M. Bressan, F. Fiechter *et al.*, “Image dataset of disk diffusion assay scanned with the SIRscan system,” Dryad Dataset, University of Zurich, 2024.

CHAPTER A: แบบสอบถามการประเมินระบบ

แบบสอบถามที่ใช้ในการประเมินระบบมีหัวข้อดังนี้:

ส่วนที่ 1 — ข้อมูลทั่วไปของผู้ทดสอบ

- ตำแหน่ง/บทบาท
- ระดับประสบการณ์กับการอ่านผล AST (Disk Diffusion)

ส่วนที่ 2 — การทดสอบตามสถานการณ์ (Scenario-Based Test)

1. การอัปโหลดภาพและสั่งรันระบบวิเคราะห์
2. การแก้ไขผลลัพธ์ด้วยตนเอง (Manual Correction)
3. การดูประวัติและดาวน์โหลดรายงาน

ส่วนที่ 3 — คะแนนความพึงพอใจ (Likert Scale 1–5)

1. ขั้นตอนการอัปโหลดและวิเคราะห์ทำได้ง่ายและไม่ซับซ้อน
2. ตารางแสดงผลจัดวางได้เป็นระเบียบ อ่านข้อมูลได้ชัดเจน
3. ฟังก์ชัน Manual Correction ใช้งานสะดวก
4. ระบบช่วยลดเวลาและภาระงานเมื่อเทียบกับการวัดด้วยไม้บรรทัด
5. มั่นใจในความถูกต้องของการคำนวณและการแปลผล
6. โดยภาพรวมระบบมีประโยชน์และพร้อมใช้งานจริงในห้องปฏิบัติการ

ส่วนที่ 4 — ความคิดเห็นเพิ่มเติม

- จุดเด่น/สิ่งที่ชอบในระบบ
- ปัญหาที่พบ/สิ่งที่ทำให้หงุดหงิด
- ฟีเจอร์เพิ่มเติมที่ต้องการ
- ข้อเสนอแนะอื่นๆ